

SoK: Self-Generated Nudes over Private Chats: How Can Technology Contribute to a Safer Sexting?

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Abstract—More and more people take advantage of mobile apps to strike up relationships and casual contacts. This sometimes results in the sharing of self-generated nudes. While this opens a way for sexual exploration, it also raises concerns. In this paper, we review existing technology-assisted permissive proposals/features that provide security, privacy or accountability benefits when sharing nudes online. To do so, we performed a systematic literature review combing through 10,026 search results and cross-references, and we identified real-world solutions by surveying OS features and 52 dating, messaging and social network apps. We systematized knowledge by defining a sexting threat model, deriving a taxonomy of the proposals/features, discussing some of their shortcomings, organizing privacy-related concepts, and providing take-aways with some directions for future research and development. Our study found a very diverse ecosystem of academic proposals and app features, showing that safer sexting goes far beyond nude detection. None of the techniques represents the ultimate solution for all threats, but each contributes to a safer sexting in a different way.¹

1. Introduction

Worldwide smartphone penetration has been growing over the past years, with estimates foreseeing this trend to continue [2]. We are increasingly dependent on smartphone technology [3], [4]. As a particular case, online dating is a space where relationships are established, with a prevalence of 30% of US adults having used a dating site or app [5]. In this and other contexts, self generated nudes may be exchanged. A meta-analysis [6] found a 30% prevalence of sending nude, semi-nude or sexually suggestive pictures among young adults. In a US survey [7], 15% of girls and 9% of boys aged 13-17 shared nude or near-nude images.

Sending or receiving sexually explicit images, videos or messages through electronic means (also called ‘sexting’) does raise concerns. People who unethically use pornographic material take advantage of the anonymity, lack of geographical boundaries, unrestricted availability of software tools, and reproducibility of digital data facilitated by the Internet and new technologies [8]. Even when private pictures are never eventually misused, the act of sharing can

originate stress to the sender because of potential privacy breaches in the future. In extreme cases, cyberbullying and sextortion over nudes exchanged online has ended up with suicide in teens [9], [10], [11]. Children (humans younger than 18 years old) are arguably the most vulnerable in the face of online risks. This has motivated extensive research for children protection [12], pivoting around the idea of filtering the sexually explicit contents that they have access to or that they share with others, as a strategy to preserve their privacy and safety.

Yet, sexuality is at the core of human nature. It is a process of body and mind (linked to a variety of emotions, feelings, thoughts, and actions), as well as a socio-cultural phenomenon. Its manifestations may vary enormously from individual to individual and over the course of one’s lifetime, ranging from people who intensely experience desire since the early stages of youth, to people who never experience attraction. As Franco et al. [13] remind us, “*sexuality is an important part of the life experience and well-being of humans. It is included among the physiological needs in the well known Maslow’s hierarchy of needs.*” Bearing such considerations in mind, the Internet and new technologies can also be turned to one’s advantage, by using dating, messaging, and social network apps as a support for benign disinhibition [14], personal development [15], and self reaffirmation of the own identity [16].

This duality of competing interests produces a persistent tension between individual’s needs for both secrecy and disclosure [17]. Privacy lies precisely at the confluence of those competing needs (in the sense of ‘sharing in a controlled way’). Such idea can be correlated with the privacy calculus model [18] and its extensions [19], [20]. So rather than an inherently harmful activity, sexting should be regarded as something that has risks and that has opportunities [21].

From this perspective, we perform a comprehensive review of technology-assisted permissive approaches (i.e., techniques, methods, strategies, systems, mechanisms, proposals, solutions, features) that provide security, privacy, or accountability to the users who send or receive nudes over the chats of mobile apps. We do so with the goal of creating a reference material for researchers, policy-makers, software developers and other professionals. This knowledge can be useful for analyzing trends, identifying research gaps, conducting related studies, defining policies, and developing

1. This is an extended version of the paper in IEEE S&P ’25 [1].

new features or new app platforms. Our work is also a call to join efforts towards providing a safer sexting for everybody.

Our contributions are: (1) we build a sexting threat model from prior literature, (2) we perform a systematic literature review, merged with a survey of app features, of approaches that provide security, privacy or accountability benefits to sexters, (3) we define a taxonomy of such approaches around the threat model, (4) we develop privacy concepts emerging from that taxonomy, and (5) we suggest directions for future research and industry development.

2. The importance of sexting

Why should we focus only on nudes and dating apps, rather than on private pictures in general? We argue that nudes are among the most sensitive kind of media exchanged over private chats, representing a very challenging use case, if not the most. Nevertheless, other types of private pictures, such as passport photos and copyrighted material, also demand strong protections. We identify below several properties of auto pornography and dating apps. The list is not exhaustive, and the properties are likely not exclusive to sexting. However, our hope is that this brief analysis helps explain why sexting is such an important topic to be studied.

2.1. Nudes

Emotions involved. There may be strong feelings and emotions involved in sexting, either pleasant, unpleasant or neutral. In some cases, having distress after sexting derives from an abuse, while in other cases it does not (e.g., being rejected). Therefore, we expect the educational, privacy-enhancing, and abuse-response approaches to be complex.

Dynamism of roles. When sending sensitive information to a journalist in the context of whistle-blowing, there is a clear role of who is the sender and who is the receptor of the information. In sexting, such roles can be dynamic. For instance, participant A sends a sensual picture, then B does the same, then A sends a more explicit picture, and so on. We expect to find privacy features adapted to this dynamism of roles.

Consensuality of exchanges. Sending a passport photo or spontaneously sharing a regular selfie does not generally raise concerns for recipients, while sending nudes without consent has different implications.

Risks and safety concerns. Sexting may entail specific harm scenarios (e.g., the so-called ‘revenge porn’), and raises safety concerns about in-person contacts. The risks are higher in specific communities. Therefore, we might find approaches tailored to such communities.

Privacy of ownership information. In copyrighted productions there is generally the desire to make evident—and as much as possible—what is the real identity of the image owner, whereas sexters usually want to keep this information private. Additionally, the context is already sensitive information, which happens, for instance, when it

is visible that a screenshot has been taken from a fetish-oriented app or an LGBTQ+ app, or when the identity of the recipient is revealed despite not appearing in the picture.

Exploitability of redacted material. When distributing copyrighted movies, masking half of the face of the characters may make the material unusable for illegal sale (albeit still being a potential copyright infringement). In nudes, as long as the person is recognisable—or as long as the concerned person believes so—the material is still exploitable for sextortion.

2.2. Dating apps

Unique features. We expect nudes to be often exchanged within dating apps, as they provide a context for sexting. It might be the case that some privacy features are found in dating apps but not in other kinds of apps. Therefore, our review should cover dating apps in depth.

Anonymous interactions. When sending nudes over a dating app, authenticating the other party is not necessarily desired. Some users may wish to identify the other party so as to prevent interaction with people they already know, as a safety precaution, or otherwise. In contrast, other users may prefer keeping online interactions anonymous, as it facilitates disinhibition and provides more privacy. Therefore, we expect some of the security and privacy features of dating apps not to be based on authentication among end users.

Building trust from scratch. The whole purpose of dating apps is to meet people one has never met before, and there is usually no prior reference about the trustworthiness of the other users. In contrast, social networks are often used to connect with people one has already interacted with in person, or who have a traceable connection. At some point of the conversation, users may decide to hop from the dating app to a non-dating app. This can be because the dating app does not allow to attach pictures in chats, missing privacy features such as E2EE (End-to-End Encryption), or otherwise. In such cases, trust is bootstrapped on the dating app interaction. We expect dating apps to provide features for assessing the trustworthiness of other users.

3. Related work

Thomas et al. [22] conducted a comprehensive survey on online abuse and provide a taxonomy for reasoning about online hate and harassment. Though not exclusive to sexting contexts, it covers sexting-related abuses. Hartikainen et al. [23] studied the advice and support that adolescents receive from peers regarding online sexual risks, based on an analysis of online posts and comments.

Geeng et al. [24] surveyed the concerns of adults when sending or receiving sexts, while reporting on popular mitigation strategies. Qin et al. [25] conducted semi-structured interviews to sexters, either in commercial or recreational purposes, reporting on mitigation strategies. Coduto and McDonald [26] conducted a survey about the digital management of shared sexual content after a breakup.

McDonald et al. [27] conducted a survey and semi-structured interviews about sex workers’ risks, strategies and goals in online activities (including sexting), reporting on mitigation approaches. Hamilton et al. [28] used three case studies to show the mechanisms of harm to sex workers, and discussed future research opportunities that would benefit citizens as a whole. Gröber et al. [29] conducted a semi-structured interview study with 23 Pakistani content creators, reporting on threat experiences and security practices.

Shetty et al. [30] tested a PiTM (Person-in-The-Middle) attack against seven popular Android dating apps. Barros et al. conducted a forensic analysis of Bumble [31]. Hayes and Snow [32] analyzed security issues in dating apps and studied how they might share sensitive information with third parties. Stoicescu et al. [17] studied privacy concerns of dating apps, based on prior work and the use of two apps.

In summary, analyses of online posts, user surveys, and interviews surface harm scenarios, needs, recommendations, and popular strategies towards a safer sexting. Besides, technical surveys of dating apps report on security and privacy issues found in apps commonly used for sexting. However, to the best of our knowledge, there is a lack of literature attempting to cover technology-assisted strategies towards a safer sexting from a holistic, universal standpoint.

4. Methodology

Our methodology is composed of two main steps. Firstly, we draw from the prior literature of §3 to define our in-scope harms and to build a sexting threat model. Secondly, based on that threat model, we perform a systematic literature review, plus a review of app features, to build a taxonomy. To perform such review, we gathered public documents (academic articles, user guides, and technical references) explaining technology-assisted permissive approaches that can provide a safer sexting. To guide our review, we formulated the research questions of Table 1.

TABLE 1. RESEARCH QUESTIONS

Research Question	Motivation
RQ1 What technology-assisted methods provide security, privacy or accountability to sexters?	This question is the attraction pole to capture documents explaining proposals/solutions.
RQ2 What properties around the threat model do they exhibit?	Identifying commonalities to build a taxonomy.
RQ3 What secondary studies review a subset of such methods?	Reviews and surveys serve as connection points to reach other papers, and report on harms, attacks and mitigations.
RQ4 What app/user surveys exist on such methods?	

4.1. Scope and study scenario

We draw from the prior literature of §3 to define the **in-scope harms** that may arise from sexting. For simplicity, we only considered the harms that affect either the sender or the recipient. Namely:

Ω1. Being pressured to share nudes or reproached about the own decisions on sexting [23].

- Ω2. Sender’s will not to screenshot, retain, or otherwise make a copy of the material is disrespected [24], [29].
- Ω3. Sextortion (financially motivated sexual extortion), or blackmailing over nudes [22], [24], [27], [29].
- Ω4. Cyberbullying (using technology to harass, threaten, embarrass, or target someone) [17], [22], [24], [29].
- Ω5. Doxxing (publicly providing personally identifiable information online without consent), such as in publicly revealing the real identity of a faceless nude [17], [22], [27], [28], [29].
- Ω6. A user device, other than sender’s or recipient’s, gets a copy or an external capture of the nudes (non-consensually, but not necessarily from intentional behavior) [17], [22], [23], [24], [25], [28], [29]. This includes the NDII (non-consensual disclosure of intimate images), as in the so-called ‘revenge porn’.
- Ω7. A person, other than the sender or the recipient, sees the nudes (non-consensually, but not necessarily from intentional behavior) [24], [26].
- Ω8. Self-harm: the sender leaks information that was not meant to be shared [24], [25].
- Ω9. Being fooled when receiving non-authentic or deceptive nudes [22], [24].
- Ω10. Receiving unwanted nudes [22], [23], [24], [29].
- Ω11. Nudes (or derived information) are covertly exploited, stored, or shared with other companies for monetization, analytics, or other purposes [17], [24], [25], [32].

Based on that same literature and the above list of harms, we built in Table 2 a sexting threat model by collecting specific attacks/incidents through which harms effectively occur. We also identified the actors, plus generic threats that connect each actor with the corresponding attack/incident. The **in-scope threats, attacks and incidents** of our study are precisely the ones outlined in that table.

Finally, to avoid being potentially faced with an unmanageable diversity of situations, we simplified our study scenario by narrowing it to the **sexting contexts** that satisfy all of the assumptions below:

- Despite being susceptible to attacks, the OS platform (both on server and on device side) is trusted.
- The app platform² is either *honest* (no attack or privacy breach is attempted) or *semi-honest* (while providing all expected functionality, it covertly exploits image data). Either case, app platforms may have vulnerabilities.
- Only pictures are exchanged. We therefore skip approaches that work exclusively with audio, video or text formats. We focus on pictures because they are significantly more prevalent than video in sexting [24].
- The pictures are exchanged using a mobile phone. We restrict to this scenario because mobile phones typically have an integrated camera, and they are used pervasively in sexting [7], [33]; in fact, more than computers or tablets [24]. We focus on Android and iOS mobile OSes because, combined, they represent more than 99% of worldwide market share [34].

2. Within ‘app platform’ we include: app binary on sender and recipient devices, servers, and sponsors/collaborators.

TABLE 2. IN-SCOPE THREATS, ATTACKS, AND INCIDENTS

Actor	Generic threat	Specific attacks and incidents	Harms
Recipient	A. Interfering with someone's will	A1. Gaining the trust of someone to then put pressure to share nudes or make the other person feel bad for not doing so (cyber grooming) [23]	Ω_1
	B. Making unauthorized copies	B1. Taking a screenshot, or doing screen recording with a system feature or a third party app [24] B2. Screen-cam (taking a picture of a screen from another device) [24] B3. Hacking on the own device or the own communications to bypass restrictions controlled by the app or the OS [23], [24], to then retain/extract unauthorized copies of the material	Ω_2
	C. Spreading information to others or threatening to	C1. Uploading the material to a public post or to a profile, forwarding to other users, exploiting it, or revealing the identity (or threatening to do one of such things) [17], [22], [23], [24], [25], [27], [28], [29]	Ω_{3-6}
Sender	D. Leaking information that was not meant to be shared	D1. Own personal information is leaked through image metadata, e.g., geolocation data [24], [25] D2. Deanonymization, e.g., revealing own identity through unique body features such as tatoos [24] D3. Sending the wrong picture or to the wrong person by mistake [24]	Ω_8
	E. Sending non-authentic or deceptive sexual content	E1. Sexual deepfakes [22], [29] E2. Completely synthetic pornography [22] E3. Pictures that are not of the sender [22], [24] E4. Nudes that misrepresent sender's age or physical appearance [17]	Ω_9
	F. Sharing nudes without recipient's consent	F1. Sending unwanted nudes during a conversation [22], [23], [24] F2. Cyberflashing (sending nudes to unsuspecting strangers, typically via Bluetooth or AirDrop, or as the first chat message from an unknown contact) [29]	Ω_{10}
App platform	G. Covertly sharing with other companies, exploiting or storing information	G1. Exploiting the transmitted nudes or information derived from them [24], [17], [25], [32] G2. Scanning the entire phone gallery [24] G3. A semi-honest app sponsor or collaborator of a honest app platform company has malicious code in the embedded or dynamic ads [32]	Ω_{11}
Third parties	H. Someone accesses the physical device, and gets/sees the nudes	H1. Phone gallery manual inspection by coercion of the victim to unlock the device, access recovery procedures, or known/trivial credentials [22] H2. Phone gallery monitoring by installing stalkerware on the victim's device [22] H3. Obtaining personal data locally with forensic or hacking tools [24], [25], [28], [32] H4. Children inadvertently see the sent/received nudes when parents lend them their phone [26]	$\Omega_{6,7}$
	I. A network adversary targets in-transit data	I1. Eavesdropping on HTTP connections [32] I2. Using a falsified digital certificate in a PiTM attack [30]	Ω_6
	J. A remote attacker targets the servers	J1. Exploiting a bug or vulnerability on the app server [24] J2. Accessing a remote system backup or cloud-synchronized data of a former partner [22]	Ω_6
	K. The user screen is seen/captured by surrounding individuals/devices	K1. Shoulder surfing (spying on the user's device by looking over the victim's shoulder in order to obtain private information) [24], [25] K2. A bystander unintentionally sees the user screen [24] K3. A nearby phone or CCTV camera externally captures the screen (third party screen-cam)	$\Omega_{6,7}$

- A dating, messaging or social network app is used. Evidence from [24] suggests that SMS and third-party apps are the main means of sexting in mobile, rather than browsers, since the web-based extensions of Snapchat, Whatsapp and Facebook Messenger are only available on computers. We note that neither Google's or Apple's stores allow commercial sex apps [35].
- The pictures are exchanged in a 1-to-1 chat. We skip approaches specifically designed for multiparty conversations, or only applicable to public or semi-public posts on social media. Evidence from [24] shows that DMs (Direct Messages) are the main means of sexting.
- Only one person appears in the pictures, and that person is the sender (or at least so is expected by recipients). We therefore skip approaches designed to tackle third-party privacy issues when sharing multiparty pictures.

4.2. Inclusion-exclusion criteria

We included approaches:

- Theoretical, prototyped, or real-world.

- Software- or hardware-based.
- Deemed directly or potentially applicable to our study scenario, even if their use case is unspecific or differs from ours.
- Published anytime, i.e., we did not apply restrictions on the publication date.
- (Within our surveyed apps) that help assess the trustworthiness/veracity of other profiles or that allow to control the visibility/reachability of the own profile by other users, even when the app does not have the feature of sharing pictures in the chats.
- Without applying a score bar for publication venues. Nonetheless, for papers from weaker venues, an additional effort was made to distill the relevance of the work.

We excluded approaches:

- Not leveraging any technology or technical means to achieve their purpose.
- Restrictive in nature, i.e., not allowing the exchange of self-generated nudes between two people who want to.
- That do not clearly define their system proposal.

- Not providing any tangible or plausible value.
- With an ambiguous, vague or inconsistent threat model.
- About availability threats, such as data erasure or service disruptions.
- Dealing with image data integrity attacks performed by the app platform.
- Where the sender is profiled as a hacker, for example by spreading viruses through the shared image files.
- Not officially documented in the English language.

4.3. Search strategy for papers

Our review is based on Kitchenham et al. guidelines [36]. We crafted 113 search strings, which are listed in Appendix A. These were derived from our threat model and protection strategies reported in the related work of §3, as well as our own experience, prior knowledge, and speculated strategies. Not all search strings have sexting-related keywords, so as to capture applicable approaches whose motivating use case is unspecific or differs from ours. We used Google Scholar given its comprehensive coverage [37]. The first 50 results of each search query were checked.

We then selected the papers whose title suggested a possible fit. Not only did we check system proposals (RQ1), but also related reviews and surveys (RQ3 and RQ4). From those, we read the abstracts to refine our selection. We finally read the full text of the remaining, to determine final inclusion based on the fit to our study scenario as defined in §4.1, and the inclusion-exclusion criteria defined in §4.2.

Next, we checked the backward and forward references of our selected papers and of the related reviews and surveys, to add the ones presenting novel approaches (snowball technique). However, we did not further check any references on algorithms, as they would add a large amount of proposals not representing new strategies for the issue. We also omitted new references about software integrity or vulnerability detection, as these are less central to our research problem. We repeated the process iteratively for any newly included papers. Appendix B contains the resulting list of 138 articles explaining some approach.

4.4. Survey strategy for app features

We considered the top five apps in the Social, Dating and Communication/Messaging categories of Android and iOS, the top results when looking up “gay dating”, “women dating”, “lesbian dating” and “trans dating” in both platforms’ official app repository for Portugal, plus other apps referenced in papers we came across, including faith-based apps. Popular sexting apps were covered³ [24], [26], [27], [33], [38]. We did not find any mobile app for sex workers other than the Hunqz section of Romeo⁴, though. In total, 47 apps were analyzed (26 dating, 13 messaging, 8 social

network), which are listed in Appendix C. We note that 9 of them (8 dating, 1 social network) do not support the sharing of pictures in chats (except for stickers).

We analyzed the apps from an end-user perspective with the aim of identifying relevant features, using the same inclusion-exclusion criteria as with the papers, and checking the corresponding online documentation. To do so, we followed the procedure explained in Appendix D. We did a similar exercise at the OS level with Android and iOS, including the native Messages/SMS app, reviewing APIs and features that we thought could relate to our study scenario. Appendix F lists the 81 in-use tools, solutions and app features that we have identified.

4.5. Data collection procedures

When analyzing proposals and features, we made every effort to assess how they can provide value in our study scenario. In other words, we evaluated their applicability. We merged the selected documents from §4.3 and §4.4 (web pages and articles) into a unified table. In almost all cases, each document explains one approach. For each approach, we evaluated and annotated a number of properties, including the ones in Appendix E.

To see how many papers “mention sexting- or dating-related scenarios as a use case” or “mention private messaging as a use case or a concern” (see §7.5) we used `pdfgrep` with keywords. To check cross-references we mainly relied on scholarly database filters. To locate documentation of app features, we performed queries such as “disappearing messages site:messagingapp.com” in a web search engine.

The procedures were iteratively refined, re-extracting data whenever necessary. For instance, more app searches and search strings were added from authors’ discussions and external reviews. All authors were involved in the whole process and held discussions in meetings. Additional analyses to items were made, especially to determine inclusion, and fit to each of the threats in our model.

5. Findings

Based on the threat model outlined in Table 2, we built a new taxonomy of approaches. Its chosen categories are derived directly from the generic threats, except in some cases where we will justify. For convenience, we manually cluster examples of approaches within each category, grouping them by similarity of the overall strategy. Note that we provide illustrative examples, but not all of the selected papers are cited. While we occasionally mention apps in our examples for the sake of clarity, we do not make any recommendation towards any platform or vendor. Although we provide some insights about the benefits and drawbacks of the approaches, our taxonomy reads as an overview, as we do not perform a comprehensive comparative analysis. Figure 1 illustrates the universe of approaches in a diagram.

We start off with features, offered by the software platform, to protect the privacy of the sender in the face of recipients. Instead of distinguishing between threats *A,B,C,D*,

3. Bumble, Facebook Messenger, Grindr, Hinge, Instagram, Signal, Snapchat, Telegram, TikTok, Tinder, WhatsApp, and X/Twitter.

4. The Hunqz section of Romeo is not available on the app but Hunqz conversations initiated from the website can be continued in the app.

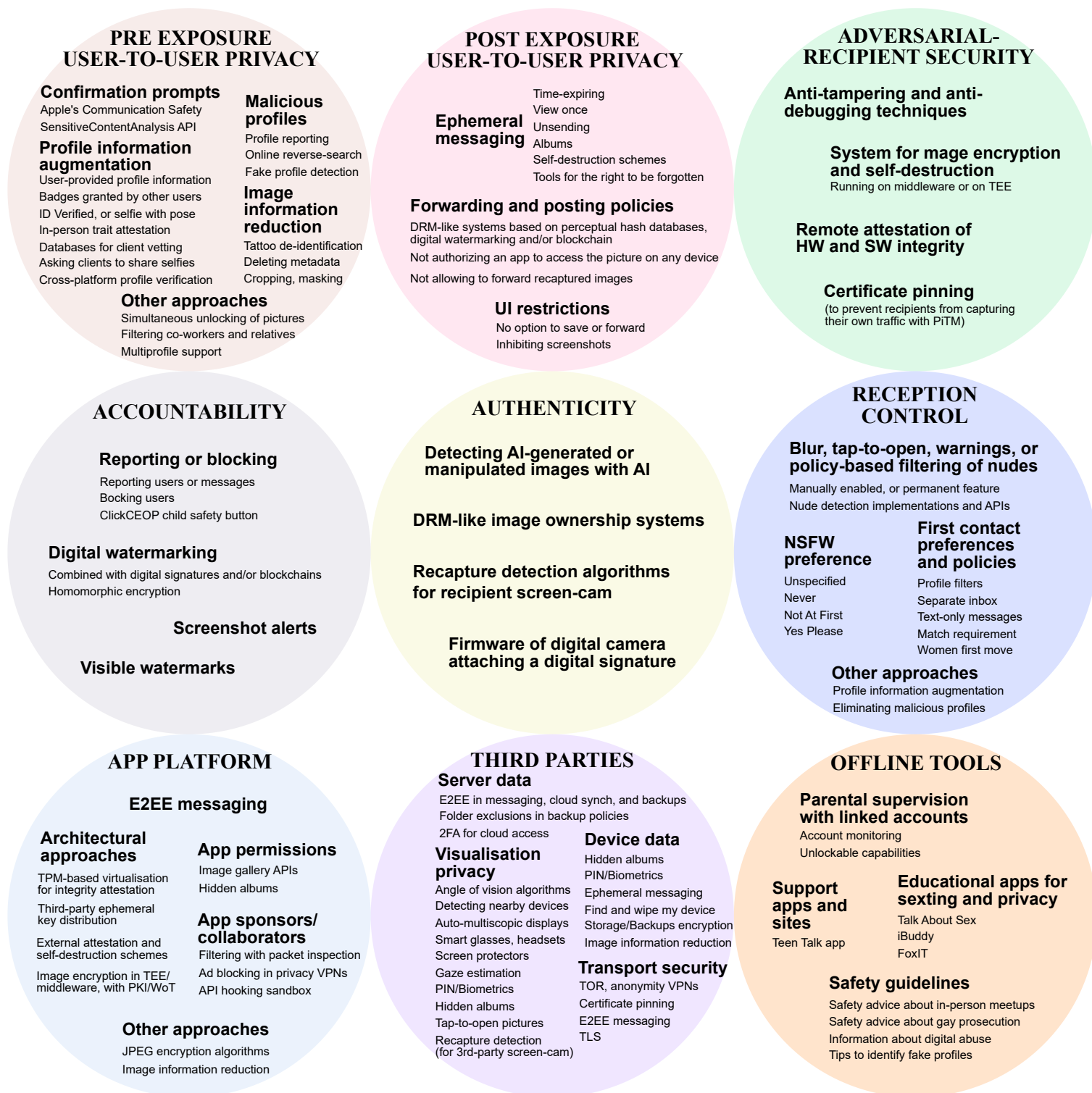


Figure 1. The universe of safer sexting approaches, organized by category as per our taxonomy.

we merge them into the same category because the attacks can be escalated on top of others. We instead discriminate between mechanisms that are enforced before (PRE EXPOSURE USER-TO-USER PRIVACY, §5.1) or after (POST EXPOSURE USER-TO-USER PRIVACY, §5.2) image delivery, to recipients who lack advanced computer skills. In ADVERSARIAL-RECIPIENT SECURITY (§5.3) we focus on recipients with hacking skills who attempt to bypass privacy protections (attack *B3*). This particular organization allows to make the observation that the security of Post Exposure User-to-User Privacy highly depends on Adversarial-Recipient Security protections. We finally consider mechanisms that provide ACCOUNTABILITY (§5.4) of actual or potential privacy violations related to threats *B,C*.

Next, we cover the threats derived from the sender's actions that impact or relate to the privacy of the recipient. In AUTHENTICITY (§5.5), related to threat *E*, we cover approaches that try to certify that the picture was taken by a camera, that it actually depicts the sender, and that its representation is faithful. In RECEPTION CONTROL (§5.6) we explore the situation where the user receives nudes without consent, which corresponds to threat *F*.

The app used for the sexting exchange is a central component of our study scenario. Yet, from the user perspective, there is no real desire that it gains access to the nudes, as long as all functionality is delivered. In APP PLATFORM (§5.7) we capture approaches where the app platform in use is modeled as semi-honest (threat *G*).

As the name implies, in THIRD PARTIES (§5.8) we deal with external actors. This covers individuals who inadvertently see the nudes on a lent device, monitor the user's phone gallery, gain access to personal cloud data, or manipulate an unattended, stolen, lost or seized device (threat *H*); network adversaries intercepting in-transit data (threat *I*); remote attacks on vulnerable servers (threat *J*); and visualisation privacy issues, where bystanders or nearby cameras see or capture the phone screen (threat *K*). We cluster those threats into a single category because they are largely unspecific to sexting.

Last but not least, in OFFLINE TOOLS (§5.9) we deal with informational, supervision, and educational strategies not tied to a particular sexting instance. We put them in a separate category because of their general purpose, as they address a broad range of security and privacy aspects.

Finally, throughout our study we identified techniques and approaches overlapping more than one category or threat. They are summarized in Appendix G, just for reader's convenience.

5.1. Pre Exposure User-to-User Privacy

In this category, we capture the approaches that satisfy the following two conditions. Firstly, they must address sender's privacy in the face of the recipient user (who we assume lacks advanced computer skills), which corresponds to threats *A,B,C,D*. And secondly, they must be 'pre exposure', in the sense that the intended mechanism must take effect

before delivering the image to the recipient. Therefore, its benefits are enjoyed *before the risk is taken*.

The approaches that satisfy the above conditions generally take the app platform as a trusted entity, precisely because the mechanisms are typically controlled by the app itself. The overall strategies are: (1) asking for confirmation before transmitting the picture, (2) reducing the amount of exposed information, or (3) preventing interactions with non-desired or malicious users from happening in the first place. This leads to mechanisms such as editing the image, providing reliable and rich profile information, or using apps that proactively eliminate malicious profiles.

Confirmation prompts. Within Apple's Communication Safety [39], there is a feature for the Messages app in which adolescents aged 13-17 are given several confirmation prompts before sharing pictures containing nudity, offering them the possibility of involving their parents or guardians. Apple additionally provides the SensitiveContentAnalysis API framework for app developers. Both perform on-device nude detection [40], [41], which provides more security and privacy. Although the language and the interventions have to be adapted to the age of the user, confirmation prompts can be beneficial for everybody. It invites the sender to reflect whether sexting is being done freely or under pressure (attack *A1*), it raises awareness about considering the use of alternative privacy strategies or features, it helps avoid slip-ups (incident *D3*), and it is also a reminder that the other party must have given consent (threat *F*).

Image information reduction. We can reduce the amount of information that is exposed by taking faceless pictures, cropping, painting or pixelating the face, with tattoo de-identification (incident *D2*) [42], or by removing scene details, among other tricks [43]. Anonymisation of sexts is actually common practise [24], [44]. Both Android and iOS have built-in image editing tools that allow to crop or paint the face, so that the information in the face region can be eliminated. Telegram provides the blur tool [45] to redact sensitive data or photobombers⁵. Nevertheless, blurred faces still provide information in machine-learning person recognition systems from a database of faces, compared to painted faces [46] or random guessing [47]. When applied to obscured images with some filter, image restoration techniques can also improve the effectiveness of face recognition algorithms [48]. In terms of security considerations, a bug in the built-in Android and Windows snipping and markup tool was discovered [49], which might allow to recover the original from a cropped image.

Tattoos represent a particular case since they can reveal the identity of a person (incident *D2*). Marcetic et al. [42] present a system for tattoo de-identification in still images. Their algorithm detects potential regions of the picture with the presence of a tattoo, and applies a median filter to preserve naturalness. However, unless explicitly notified, de-

5. A person who prevents a photograph from being taken in the way intended by suddenly appearing or doing something unexpected as the picture is taken.

identification techniques do not make it evident for recipients that the image was edited.

The picture itself is not the only information that an image file holds (incident *D1*). JPEG and other formats can contain extra information as EXIF data, such as GPS coordinates. The metadata is not visible, but there exist mobile apps to view and strip it [50]. There is evidence that in Signal and Bumble EXIF metadata is stripped from the images [24], [31].

Profile information augmentation. Because dating apps are meant to connect with strangers, the information of user profiles—provided that it is reliable—helps users determine whether a potential recipient is trustworthy and of interest. According to [51], older users are more concerned about the veracity of age and relationship status in dating profiles, whereas younger users are more concerned about accurate and recent photos being present in the profile.

Apps may show the number of followers, number of friends in common, user’s friends, and date joined (or ‘member since’). They may also have a Verified Profile badge on profiles that have been certified as belonging to a real person, either by using other users’ input, or by checking a fresh selfie/video with a random pose/movement against the profile pictures. In the case of Romeo, the feature relies on users voluntarily marking others as *I know this user personally* within the app [52]. Bumble’s Verified Profile feature asks users to take a fresh selfie mimicking a random pose, and, using a combination of human and automated review processes, it verifies that it corresponds to the same person as the one appearing in the profile pictures [53]. In Muzz (a Muslim dating and marriage app), users can earn ID Verified badges, plus good-behavior badges granted by other users [54]: It shows a badge after passing an ID verification process that requires to scan an official document, and users can earn badges from other users’ reviews: *Serious*, *Polite*, *Recommended* [54]. In Dinder Club (a dating and planning app for people with intellectual disabilities), only identified users who belong to participating social entities can join, which effectively attests that all app users have some intellectual disability [55]. Finally, sex workers may use external services/databases for client checking and vetting [56], [57]. Asking potential clients to share several selfies is also a verification strategy used by sex workers [58].

Dating apps may allow to connect the user profile with the posts of an existing social media account. Because the user needs to log in correctly into the social media account to complete the process, this prevents false cross-platform connections. Similarly, Norcie et al.’s academic proposal [51] allows the back-ends of dating apps to get, via API, existing profile information in other social networks in order to contrast the user-provided information in the dating app profile. This is precisely the idea of Certifeye [51], which checks the user-provided information of relationship status, age, and photos against their Facebook account via API, without revealing the social network usernames among users. If the information matches, the user obtains a certification badge in the dating profile, which has to be renewed periodically. This also allows to show the actual number of

Facebook friends of the user. But aside from that, and aside from the procedures (such as credit card verification) and AI (Artificial Intelligence) algorithms aimed at verifying ≥ 18 in dating apps and ≥ 13 in social networks [59], [60], [61], there is currently no reliable way to ensure the veracity of the age stated in the profile.

Malicious profiles. App platforms routinely purge the userbase from fake profiles, which helps combat bots and users who collect nudes for unethical purposes. In the industry, fake profiles are detected with a combination of automated tools and human intervention, with AI [62], or even with the collaboration of users in profile reporting [63]. In the academy, proposals are based on graph-modelling of social networks [64], UI behavioral pattern analysis [65], Long Short-Term Memory neural networks [62], or reverse-searching of profile pictures online [66]. This explained more in detail below.

One approach for detecting fake profiles is to model the social network as a graph. Viswanath et al. [64] studied several existing graph-based Sybil defenses to show that they are implicitly based on the ranking, that is, the number of links to trusted nodes. They found that the natural presence of sparse, poorly interconnected communities within social networks diminishes the effectiveness of such defenses.

Taking a different road, Wang et al. [65] made the observation that because Sybils have different interests than legitimate users, they focus on matching more friends and disseminating messages, whereas real users partake in a broader range of activities. To exploit this idea, they use server-side clickstream models to group behavioral patterns, and classify users with that information. More recently, DatingSec [62] developed a long short-term memory neural network, plus an attentive module, that uses static features (profile and community information) as well as dynamic features (textual information of posts and comments, post time, number of views, etc.) to detect malicious accounts in dating apps.

To combat scammer profiles, Social-Guard [66] suggests comparing the profile pictures with images of celebrities, and reverse searching the pictures online to see if an already existing image matches, by using AWS Rekognition and Google Vision APIs. In sextortion, scammers may gain the trust of a victim to mutually exchange intimate pictures, and later the victim receives a message of someone claiming to be the parent of a minor who received sexually explicit photos, threatening to report to the police unless some amount of money is paid [66]. Within the US, the possession of child pornography is illegal, even when it is self generated. In fact, minors have been charged when sending self-generated nudes or when in possession of nudes of other minors [67], [68], [69]. Therefore, in such jurisdictions, children can also be target of child pornography sextortion.

Other approaches. The League (a dating app) allows to link the user profile with LinkedIn so as to prevent matches with co-workers [70]. This strategy helps avoid potentially embarrassing situations in which users engage in anonymous sexting to then discover to be relatives or work colleagues.

Who sends the picture first is a common concern among users [71]. Although we have not found any implementation, users demand a feature to ‘trade pics’ in a way that the information is simultaneously unlocked on both ends, enabling more egalitarian conditions of intimacy self-disclosure [72]. In *mutual privacy*, as we call it, the ability to control the flow of information is exercised reciprocally, and in an egalitarian way. This applies to activities that are bidirectional (such as trading pictures, or browsing each other’s profiles), as opposed to activities that are unidirectional (such as receiving but not sending pictures, or when browsing profiles in incognito mode).

Multiprofile support for social media apps currently fails to address sex workers’ needs of maintaining completely separate online personas, as current technologies are not mature and have usability issues [28], [73].

5.2. Post Exposure User-to-User Privacy

Post Exposure User-to-User Privacy is analogous to Pre, except that the mechanisms are enforced *after* image delivery (regardless of when the feature is configured or enabled). Consequently, we deal with malicious or unintentional behavior by recipients (threats *B,C*). We do not deal, though, with recipients one does not want to interact with, precisely because we are assuming that the picture has already been delivered. In general, those mechanisms provide privacy to the sender by limiting what the recipient can do with the picture, or by controlling its availability over time.

UI restrictions. The software can disable the handy features (contextual menus and intents) of saving received pictures to the phone gallery or forwarding to other users. Another strategy is to inhibit screenshots programmatically (attack *B1*). Android API can do so natively, while in iOS there are workarounds to produce an equivalent effect. In iOS, it can be disabled via MDM (Mobile Device Management) [74]). As another option, the Confide app claims that its ScreenshieldKit technology has prevented more than 7.9K screenshot attempts [75]. In Grindr, screenshots are inhibited in albums and expiring photos [76], [77], while in the Scruff dating app received photos over private chats and albums cannot be forwarded, screenshotted, or saved to the device; they can only be saved to a Scruff album for later viewing [78].

UI restriction techniques just make it more difficult for recipients to make copies of the material, since they generally cannot prevent: (1) screen-cam captures, (2) screen electronics manipulations, (3) rooting the device, (4) acquiring storage or RAM data to explore a snapshot, (5) backup analysis, (6) using a modified version of the app, (7) doing a PiTM to intercept the own data stream, or (8) taking a screenshot from an external mirroring display [79], [80].

Ephemeral messaging. Usually, messages are stored on the recipient device for as long as the recipient user wants. In ephemeral messaging (see Figure 2 for a suggested classification), the software logic of the app (either controlled by the app or by a middleware) implements policies

that limit the availability of messages over time, at sender’s discretion.

Self-destructing messages

- **Time-expiring** Content is deleted when a timer expires (usually configurable in magnitudes of minutes, hours, or days). It can be viewed an unlimited number of times, though. Available either as a chat option or on a per message basis. Apps may label it as *disappearing messages*, *auto-delete* or *time capsule*.
- **View once** The message can only be viewed once (or a very limited number of times, e.g., view twice). Typically assisted by a tap-to-open button and combined with a short timer. Apps may label it as *disappearing message* or *view once*.

User-controlled removal

- **Unsending** The deletion on the remote device, if desired, is triggered manually by the sender. Ideally, the app also triggers the deletion of the corresponding system notification on the recipient device. In the chat history, the app may explicitly signal that a message was deleted. Apps may label it as *retract*, *unsend* or *delete for everyone*.
- **Albums** This variant applies exclusively to media. The recipient is granted access to a pre-configured photo album hosted on the app servers. Access can be revoked anytime.

Figure 2. Suggested classification of ephemeral messaging features.

Ephemeral media is vulnerable to the same attack vectors we identified in UI Restrictions. In addition to intrinsic limitations, implementation issues and design caveats have been found. In the past, forensic analyses of Snapchat showed that the snaps were not completely eliminated [81], [82]. In Telegram, time-expiring images were logically deleted from the file system, but image artifacts could be recovered with forensic tools [83]. Under certain conditions, Whatsapp does not remove the images from the phone gallery when they are ‘deleted for everyone’, specifically in the cases where the recipient has an iPhone and the feature “Save to camera roll” is ON (which comes enabled by default, at the time of writing this document). Whatsapp claimed that the control of images in the phone gallery is out of the intended scope of the functionality [84]. Additionally, ‘disappearing messages’ can be recorded or forwarded to third parties before the timer expires, remaining in the third party’s chat indefinitely provided that the option of disappearing messages is not enabled in it [85].

We cite Kaboom [86] (now discontinued) as a special case, as it is not a messaging platform per se. It allowed to share ephemeral messages, including images, over any messaging platform. Users could share links to the content, and set it to self destruct after a certain time or number of views [87]. Because ephemeral messaging is commonly

provided as a built-in option in apps with a chat feature, we do not foresee a re-emergence of Kaboom’s approach.

Disparities between UI labels and documentation wording may induce to confusion. For instance, Whatsapp’s *time-expiring* is labeled and documented as “disappearing messages”, but Instagram’s *view once* is labeled as “view once” yet documented as “disappearing photos” [88], [89], even though both platforms are controlled by Meta. In Wickr’s and Signal’s time-expiring messages, the timer starts when the message is sent, or read, respectively [90], [91].

Stokes and Carlsson [92] present an academic proposal for the right to be forgotten across social networks, by combining perceptual hashing, digital watermarking, and digital signatures. This system assumes cooperative behavior among users, and images persist in plain in the app servers. When a user wants to exercise the right to be forgotten on a picture, a software agent installed on the device broadcasts a signed request to a peer-to-peer network that eventually reaches all other agents. Receivers use a combination of perceptual hashing, digital watermarking and digital signatures to validate requests. If the received request is legitimate, the agent instructs the social network client not to show the picture to the user. This, in practice, provides cross-social network digital oblivion with regards to user-to-user privacy. However, the contents are not deleted from the servers, and it assumes cooperative behavior of users by voluntarily installing the agent. Some self-destruction schemes [93], [94], [95], which we will cover in Architectural Approaches in §5.7, also provide the user-to-user privacy benefits of ephemeral messaging, in addition to protecting against semi-honest or buggy apps.

Forwarding and posting policies. In various academic proposals, the app platform enforces forwarding/posting policies established by the sender, or supports methods to prove copyright in requests for taking pictures down from websites. For policies like *only the owner can post this picture or forward it to other contacts*, academic proposals suggest using some form of ownership proof/verification similar to DRM (Digital Rights Management) systems, based on perceptual hashing [96], [97], digital watermarking [98], [99], and/or blockchain [97], [98]. Similarly, real-world initiatives such as `takeitdown.ncmec.org` or `stopncii.org` use hashing techniques to register intimate images, and prevent uploads within participating media platforms. Pictures are hashed on-device, but they have to be manually registered. We note that there are APIs for DRM in iOS and Android [100], [101], [102], in which plaintext media is not accessible from user execution space, but current implementations only support audio and video codecs.

In SafeSext [96] academic proposal, before allowing the user to send or forward a picture classified as containing nudity, a remote server calculates its perceptual hash value (using an image fingerprinting function that the authors develop) and checks it against a central database. If a match is found with a picture such that their hash distance is below a configured threshold, this means that the images are very similar, and then the forwarding policy blocks the attempt

to send the picture, if established by the policies. Otherwise, the new hash is stored to the database for future checks. The work of [97] extends this idea with two additions. Firstly, they use a Non-Transferable NFT (Non-Fungible Token) contracts instead of a central database. NFTs are sustained on a blockchain and are therefore decentralized. Secondly, they consider the execution of the image forwarding algorithm on the sender’s or the recipient’s device. This allows for more privacy, and compatibility with E2EE.

Instead of perceptual hashing, we can leverage digital watermarking (see §5.4) for ownership identification, as explained in the following two examples. In [99], a keyless invisible digital watermarking algorithm inserts plain data about the user who owns the image (a 10-digit phone number, an 8-bit identifier of the messenger application, and/or the email). The watermark should be inserted only if there isn’t any already present. This helps identify the source of an image. In this way, apps may be able to prevent the unauthorized distribution and spread of multimedia data over social networks. In Go-Sharing [98], a smart contract platform based on HyperLedger Fabric 2.0 is used to enforce owner’s and forwarders’ privacy policies over images, with a dissemination tree structure to maintain policy data for each image. Because the blockchain would be common among all app platforms, the policies can be enforced in cross social network scenarios of image re-sharing. They also use Deep Learning techniques to insert identifying information to images with invisible digital watermarking, in a way that is even robust to non-standard forms of noise.

Recapture detection algorithms [103], [104], [105] can be used to *inhibit the forwarding of pictures not taken from a live scene* (i.e., taken with screen-cam (attack B2) or from a printout). Recaptured images create artifacts and changes in texture, such as the moiré pattern in recaptured pictures from LCD screens. Recapture detection algorithms take advantage of those effects to detect the liveness of a picture, by checking features in the frequency domain, by leveraging Convolutional Neural Networks, or otherwise. However, such algorithms have been more tested in the attack vector of presenting a printed face to a facial authentication system with the intention of gaining unauthorized access. If executed server-side, they are incompatible with E2EE, as the references we found do not operate on encrypted domain.

Guo et al. [106] suggest using a middleware and a central policy server to implement policies of the style *App X is not authorized to access this image on any device*. It assumes collaborative behavior among users. A central policy server is used to store user policies (e.g., “Alice does not want app A to access her photos”). When Alice sends a photo of her to Bob using app B, the middleware installed in Bob’s device downloads the policies from the server and prevents Alice’s photo from being accessed by his app A. In this way, Bob cannot forward the photo using app A.

5.3. Adversarial-Recipient Security

The security of sender-privacy features controlled by the recipient app, as with ephemeral messaging, heavily rely on the ability to protect against recipients with hacking skills (attack *B3*). Hackers might do a PiTM [30] of their own traffic to capture the received plain data regardless of app restrictions. Alternatively, they could run a modified version of the app as a way to circumvent app restrictions, for which a number of anti debugging and anti tampering protections exist [107]. Figures about social, dating, and messaging apps’ prevalence of those protections in Android appear to be around $\pm 10\%$ from the global average [107].

Set aside proxy-based and hardware attacks, the main threat is a software integrity breach. As anti-tampering protections, we have: signature checking, code integrity checking, installer verification, and software integrity attestation. More than 90% of social and dating apps implement at least one anti-tampering protection, whereas communication apps fall a little above 80%, thereby not reaching the 90% average [107].

Hardware-backed integrity attestation on mobile platforms [108], [109], which uses a chain of trust bootstrapped on hardware, is of interest in our study scenario. With DeviceCheck in iOS [110] and Play Integrity in Android [111] (formerly SafetyNet), apps can verify from server side that a remote device runs unmodified app code. To achieve so, app integrity APIs leverage hardware-backed remote attestation—available in Android depending on the phone model [108]; undocumented in iOS although likely using iPhone’s built-in Secure Enclave [109], [112], [113]—, combined with Secure Boot [112], [114], [115] and a chain of trust mechanism. According to a 2020 study [107], which surveyed 23,610 Android apps with static analysis, 38 of our analyzed apps did not implement SafetyNet attestation or did so incorrectly (while the remaining 9 are not included in their study). Later surveys combined static and dynamic analysis, finding that, virtually, no Android app uses app integrity APIs correctly [116], [117] (where they surveyed 163,773 Android apps in 2021 combining static and dynamic analysis), and in a survey of 1,836 popular apps only 1.1% of iOS apps use it at all (either correctly or incorrectly) [117].

Attackers may also use debugging techniques to gain knowledge about the program code, to later use this knowledge to tamper with the app binary so as to alter its normal behavior. To mitigate this threat, apps may implement anti-debugging techniques, namely: emulator detection, dynamic analysis framework detection, debugger detection, debuggable status detection, and altering debugging memory structure. More than 70% of social and dating apps, and more than 60% of communication apps, implement at least one type of anti-debugging protection, which is above the 59% average [107]. In theory, app integrity APIs can even mitigate PiTM when used in conjunction with certificate pinning. However, such protections are ineffective in backup analysis, and in RAM/storage acquisition attack vectors. Furthermore, an attacker might exploit a vulnerability of

the OS to escalate privileges, so as to hack privacy features while remaining unnoticed by the remote attestation API. Since in current APIs the OS version number is not included in the attestation token, the app server cannot reliably reject OS versions that are known to be vulnerable.

With regards to academic proposals, TEE (Trusted Execution Environment) can be leveraged precisely to mitigate OS vulnerabilities [118]. Also, remote attestation systems not controlled by the app platform [119], [120] protect from adversarial apps in addition to adversarial recipients (we will cover them in Architectural Approaches in §5.7).

5.4. Accountability

In this category, we explore how technology helps answer the question ‘*who did it?*’ when a potential or actual privacy breach related to user-to-user privacy has occurred. This can be useful to take appropriate action, such as preventing the repetition of future attacks (to the own user or to others). Our findings include in-platform reporting options, screenshot alerts, and visible watermarks. As a cutting-edge line of research, digital watermarking algorithms can be combined with smart contracts and digital signatures.

Reporting and blocking. Apps typically offer the option to block users and to report violations of the platform’s community guidelines, which can help tackle sexual deep-fakes (attack *E1*), non-consensual sexting (attacks *F1*, *F2*), sextortion (attack *C1*) and other online abuses. Usually, the last exchanged messages are optionally or automatically attached to the report, even when the media is no longer visible as a result of it being unsent or self-destroyed.

Other solutions are tailored to children. In the UK, Facebook offers a child-safety button (the ClickCEOP button in their homepage) for young users, which connects to the Child Exploitation and Online Protection Centre [121].

Screenshot alerts. Some apps detect and notify users when the remote party takes a screenshot, with the caveat that no method is bullet-proof [122], [123]. Senders may then decide not to send more nudes to someone who takes screenshots, until the situation is cleared. Nevertheless, saving received sexts and taking screenshots in Snapchat is common practice [24], [26], [124].

Visible watermarks. With Android tools or third party photo editing apps, one can insert a visible watermark with a custom text [125], [126]. The watermark could be a secondary email address or an anonymous social media account. OnlyFans (a web-based content creator platform) supports visible watermarks [127], but we are not aware of any sexting mobile app that supports it natively.

Digital watermarking. Steganography and digital watermarking refer to the art of hiding information within another message or physical support in a way that its presence is not evident to human inspection. Quoting Evsutin and Dzhanashia [128], “*steganography is used to hide information from third parties while watermarking is used to add additional information related to the container.*” Digital watermarking focuses more on robustness, in the sense of making it difficult to corrupt the embedded data without

affecting picture quality. It is an area of extensive research in copyright protection applications [129], [130], [131]. In our sexting scenario, it can provide leakage traceability in case the private picture is maliciously forwarded or posted (as long as a different watermark is inserted every time the image is shared). It is therefore a means to attach metadata information about the context, particularly: app name, sender and recipient username, location, and timestamp. There have been recent advances in addressing two specific robustness challenges: resistance to common image transformations such as resizing or cropping [132], [133], and persistence after a screen-cam operation [132], [134], [135].

Watermarking algorithms normally use a symmetric secret key for watermark insertion and extraction, and, in what is known as *non-blind watermarking*, the original image is also necessary for the extraction [131]. It is assumed that the recipient does not know the key, although in some schemes the presence of a watermark can be detected without the key. Embedding capacity is another property, although in our study scenario we do not have high requirements, especially if we use the watermark only to embed a unique identifier that is referenced by an external database or blockchain. If the watermarks are *fragile*, it means that they are destroyed with any image alteration, while *semi-fragile* watermarks persist with legitimate transformations such as compression. Some watermarking schemes are *reversible*, meaning that they allow to partially or completely recover the original image from a watermarked image.

With digital watermarking, we need to involve an intermediary to certify the integrity of the process. Otherwise, the sender could insert a watermark with arbitrary information, and hold false accusations. In the DRM scheme of [136], an intermediary (which could be the app platform itself) inserts the watermark, then the involved parties digitally sign it, and the intermediary registers the digital signatures and the image hash to a blockchain. In this way, all the involved parties are able to verify the attestation. More precisely, the app would insert the watermark (having knowledge of the symmetric key), and it would provide the recipient with a token containing the relevant watermark information and a hash of the image. All the involved parties would sign the token. The hashes and the signatures (but not the watermark information or the image in the clear) could be registered in a blockchain, so that the sender and the recipient can independently verify the transaction in a public, immutable ledger.

In the approach outlined above, the app needs to access the plain image. To address this privacy issue, the fully homomorphic encryption scheme of [137] allows to insert a watermark (previously encrypted with the sender's public key) to the image ciphertext, and then register the image exchange as a transaction in a smart contract. This provides the benefit that the app does not need to know the plain image or its encryption key. However, the algorithm takes at least a few minutes in a desktop computer to insert the watermark in small grayscale images.

5.5. Authenticity

Users may share sexual deepfakes [138], in which there is a false portrayal of the intimate sphere (attack *E1*), or AI-generated images not depicting anyone in the real world (attack *E2*). Users may also share pictures they do not own, i.e., received from others, obtained with screen-cam, non-consensually taken, or downloaded from the Internet (attack *E3*). Finally, images may be deceptively altered or be decades old, ceasing to represent the user within reason (attack *E4*). The concern we address here is that the recipient may be fooled to open a space of intimacy with someone who is not truthfully doing so. Even if the picture is entirely synthetic, the recipient's privacy is affected because of expectations of egalitarian privacy (if one shares authentic pictures, one expects received pictures to be authentic too).

As for mitigations, there exist algorithms to detect image manipulations [139]. More recently, CNNs (Convolutional Neural Networks) are leveraged to perform such task, covering multiple types of manipulations [140], [141]. The use of AI itself is being explored to detect AI-generated images [142], [143], although this strategy is not generalisable to all AI architectures.

Image ownership systems and recapture detection algorithms (mentioned in Forwarding and Posting Policies in §5.2 for senders' privacy) also benefit recipients' privacy [144], [145]. For example, the perceptual hash of a received picture could be checked against a database of registered pictures containing pairs of (*perceptual hash*, *public key*). If there is a match, the sender would then be required to provide a proof of ownership of the corresponding private key. However, this would not guarantee that the person appearing on the picture is the sender.

In the industry, some cameras have a firmware that automatically attaches a digital signature to the pictures taken [146], [147]. However, they are based on manual camera registration using on-premises software in corporate environments. For our study scenario, a PKI (Public Key Infrastructure) or a permissioned blockchain would be needed for recipients to be able to verify signatures independently. Even so, attackers might be able to tamper with the firmware to register existing pictures of other people as their own.

5.6. Reception Control

Broadly speaking, 'privacy' is the "*interest that individuals have in sustaining 'personal space', free from interference by other people and organisations*" [148]. This usually refers to the right and the ability of the citizen to control the information that is exposed to others. Nevertheless, in 'reception control', as we call it, we consider the opposite direction, as it refers to controlling the information and the communications to which one is exposed.

Discord has an option to silently drop DMs with sexually explicit images received from non-friends [149], although in false positives the user is unable to send the picture [150]. Jack'd, instead, offers recipients an option to blur any restricted content until they tap the image to show it [151].

Interestingly, both features are labeled as “sensitive content filter”, although they work very differently in each case. Bumble uses AI to automatically blur nudes shared in private chats [152]. The recipient gets a notification, and can then decide to view, block or report. However, this feature cannot be disabled. In Telegram and Confide, the sender can enable an option so that the recipient has to tap the image to unblur or show it [45], [153]. Even in non-consented sexting, if the blur allows to discern that it is a nude, it might be considered that it promotes reception control.

In some dating apps, recipients can proactively signal their willingness or unwillingness to receive nudes by setting an NSFW (Not Safe for Work) preference on the profile, typically with the following options: Do Not Show [my preference], Never, Not At First, Yes Please. An NSFW preference of ‘Yes Please’ and in-conversation agreements are means of giving consent for sexting.

Other features are specifically designed to protect children while promoting parental supervision. Apple’s Communication Safety and the SensitiveContentAnalysis API work in the same way as in confirmation prompts (§5.1), before showing the received nudes. These frameworks allow to flag and blur sensitive images, and facilitate supervision interventions, before the recipient gives consent.

An alternative strategy is to have more control about the start of a chat. In swipe-based dating apps, chats can only be initiated after a match, and with Bumble’s First Move feature, only women can initiate, which may help prevent cyberflashing [154]. In Plenty of Fish, users can customize the First Contact Preferences, so that only those who satisfy specific criteria (country, age range, etc.) can initiate a chat with them [155]. However, geolocation information can be spoofed using technical means [156], and personal information on the profile can be easily misrepresented. As a way to prevent unwanted contacts, Instagram restricts DMs between children and adults who do not follow one another [61], and applies policies such as disallowing to attach images and using a separate inbox in first time contacts [157]. Going further, it has plans to roll out a feature to warn children and show reporting options when they receive a DM by an adult who has sent “a large amount of friend or message requests to people under 18”, as in Instagram’s judgment this exhibits a “potentially suspicious behavior” [61].

In our literature review we found algorithms for nudity detection, which can be used for some of the reception control interventions we covered. The techniques go from skin detection based on pixel color analysis, to more sophisticated Deep Learning techniques [158]. Aside from those algorithms for classification of plain images, there is research on encrypted-domain skin tone detection, which can provide reception control while preserving user privacy in the face of the app server [159]. Encryption may be achieved by scrambling the pixels following a reordering sequence derived from the key. In other words, the app server is able to detect nudes but not to reconstruct the original image, despite the fact that partial information is leaked. In [159], every RGB value corresponding to a skin color is randomly mapped to a range of values, so as to make

it more difficult to reverse the encryption without possessing the key.

5.7. App Platform

In this category we capture methods to protect the security and privacy of the nudes when the app platform is taken as semi-honest, e.g., it covertly exploits the transmitted nudes or the phone gallery contents for monetization, analytics, or other purposes. The mechanisms are controlled by the mobile OS, a middleware, a virtualisation layer, or an external server. To protect user privacy, the mechanisms in this category either (1) restrict the app’s access to other images, (2) employ E2EE, (3) encrypt the image file before any app component accesses it, or (4) enforce limitations on what the app platform can do with the image data.

App permissions. In 2021, a tech blogger observed that the Chinese chat apps WeChat and QQ scanned the photo gallery in the background [160]. To control the permissions of curious apps (attack *G2*), newer iOS and Android image gallery APIs restrict each app’s access to only a subset of pictures manually selected by the user [161], [162]. Furthermore, in iOS, photos in the hidden album are not accessible by apps. Such approaches reduce the attack surface of semi-honest apps, but there is still the issue that the app has access to the images the user wants to share.

End-to-end encryption. E2EE is a security and privacy feature (attacks *G1, I1, I2, J1*) in which app servers do not have access to the end user’s private conversations, as they are encrypted with a key only known to the participating endpoints [163], [164]. Facebook Messenger and Instagram have plans to make it enabled by default and available for all users [165]. Nevertheless, in some apps it has to be activated manually on a per chat basis [166], and we are not aware of any dating app that supports E2EE. Apps may offer an option for reporting abusive messages. As a caveat, when a user reports a message, its contents are sent to the app servers [167], [168]. As downsides, most apps optimistically assume that users will verify the contacts’ public keys out of band, and security is only guaranteed if the app code is verified. We finally observe that the labeling across platforms can be confusing to end users. For instance, E2EE chats are called “secret chats” in Telegram, whereas in Skype they are called “private conversations” [169], [170].

JPEG encryption algorithms. To combat semi-honest apps, specific encryption algorithms for the JPEG format encrypt images with verified keys before the app accesses them. Such algorithms provide two interesting benefits, compared to general-purpose encryption. Firstly, they produce a JPEG-compliant ciphertext [171], [172], [173]. Secondly, they have some resistance against common transformations performed by app platforms, such as downsampling, with a negligible loss of quality [172], [173]. However, this does not have a good fit with the use case of dating apps, because their main purpose is to meet people one has never met before, and users may want to interact anonymously.

The image encryption schemes have been prototyped as a browser plugin or as a network proxy, so they do

not require to make changes on the social network side. A few have been tested on a mobile platform [174]. They are commonly based on symmetric encryption, but Attribute Based Encryption is proposed in [175], [176], especially in the variant of Ciphertext-Policy, which produces a single ciphertext with and embedded policy, that can be decrypted by several recipients. Each recipient is provided with a different key, and keys have embedded attributes. This enables the possibility to grant access to new recipients even after encryption. Attribute Based Encryption can even produce JPEG-compliant ciphertexts [175].

As for security considerations, the original P3 encryption algorithm [172] has been shown to leak information that can be leveraged by deep-learning person identification systems from a database of images where the user appears [47]. Furthermore, a disadvantage of Attribute Based Encryption variants [175], [176] is that they require a trusted third party to generate and distribute users' private keys, thereby not guaranteeing their full secrecy.

Architectural approaches. In JPEG encryption algorithms we assumed that there is some secure method for exchanging keys, and then a cryptographic scheme specific to the JPEG format is defined. Architectural approaches, instead, describe how a protocol operates on top of the app, with a middleware and/or an external server, and using some pre-existing cryptographic scheme. We consider that a mechanism that does not allow the app platform to access plain image data is an approach that also takes it as adversarial, even if the corresponding paper is motivated by the use case of buggy software [95], or user-to-user privacy issues [94]. Compared to E2EE, JPEG encryption and architectural approaches have the advantage that the security of their protections persist after an app update without the need to re-verify the app code.

Some works [119], [120] propose a virtualisation layer that, with the use of a TPM (Trusted Platform Module) or some other attestation method, both on the server and on the devices, certifies the exact integrity or the semantic integrity of all the software of the app platform. With SAFE [119], the sender could check the integrity—or, rather, the compliance with what they call a SAFE environment—of the app binary. In this proposal, certified OSEs (the ones running on mobile devices and the ones running on cloud servers) use TPM to attest their authenticity against a central trusted SAFE server. Because certified OSEs are trusted, this creates an ecosystem of components linked by a chain of trust. Data is encrypted whenever it has to transit outside the SAFE environment. Certified OSEs have provisions to control how apps use sensitive data, by performing a combination of static analysis and runtime taint propagation. This proposal requires to deploy TPM hardware in the app servers and to run server code under validated OSEs. Similarly, in [120], instead of certifying that the software has not been altered in a single bit (not even a meaningless one), it provides a higher-level semantic remote attestation. It runs applications within a trusted virtual machine that ensures that certain properties of the application always hold. Modifications to binaries are tolerated as long as those modifications do not make the

program behavior violate the properties. In such proposals, though, app platforms have to accept a high level of scrutiny of their servers' hardware and software implementations, and upper layers of the virtualisation infrastructure would also have to integrate with the system.

Other proposals delete images automatically after some defined time. For example, an external trusted server independent from the social network can be used to distribute image encryption keys, destroying such keys when established by user policies [93]. Specifically, a Key-Policy Attribute Based Encryption system is proposed to enforce access constraints based on expiration time, by providing fresh keys every time the user authenticates to the trusted server and requests to access an image. However, this requires to previously authenticate users who are granted access, and it assumes that endpoints will not preserve keys. In [94], authentication is not required because recipients only have to resolve a CAPTCHA and to provide a hash of the ciphertext, yet this only mitigates large-scale attacks. The expiration time is controlled by the trusted server, and can be modified after sending the image. In both proposals, after the expiration date, neither the recipient or the app are any longer able to decrypt the image, because the key has been destroyed (provided that recipients do not preserve keys).

Proposals on sticky policies [95] and TEE [118] assume that devices are shipped with a key pair signed by the manufacturer. The private key is used by the middleware for the image encryption, but it is not accessible by the app. ShareIff [95] is a middleware service running on the mobile device, between the app and the OS, in charge of encrypting and decrypting images. It allows the sender to attach a *self-destruction policy* to images, as a form of sticky policy [177]. Then, in summary, the sending endpoint encrypts the image (and its corresponding policy) using the public key of the receiving ShareIff endpoint, presumably exchanged in-band through the app. Because the recipient public key is signed by the manufacturer, the sender can check its authenticity. Rushmore's approach [118] mitigates OS vulnerabilities but can also combat semi-honest apps. It leverages TEE to perform the image encryption and decryption operations, and to render private images on the screen in parallel to rendering non-private UI elements. TEE operates in a separate root space called *secure world*, so that if the *normal world* is compromised (e.g., the device owner gains elevated privileges), it does not affect the *secure world*. Android's TEE implementation is called Trusty, which uses ARM's TrustZone or Intel's Virtualization Technology [178]. However, Trusty is not open to app developers; only core apps can use it. The somewhat equivalent technology in iOS would be a separate subsystem in the SoC (System-On-Chip) for processing private images, but this might require hardware upgrades. Given that the remote public key has to be authenticated, we deduce that a PKI of all manufacturers or a Web of Trust needs to be deployed.

App sponsors and collaborators. Some dating apps obtain revenues with in-app advertising or cooperate with

external campaigns, in the form of banners or pop-ups. Since those advertisers are separate entities with potentially diverging interests, we have to consider approaches where the app platform company is trusted but the embedded ads are untrusted, as they could for instance scan the phone gallery covertly (attack *G3*).

Academic approaches suggest sandboxing ad library code to intercept and block sensitive API calls [179], or else to monitor network traffic with Deep Packet Inspection and TCP/IP header analysis to look for traffic features typically found in advertising traffic, to then block those connections [180]. Industry solutions, such as the ad blockers of privacy VPNs, can also achieve network containment by filtering traffic.

AdCapsule for Android [179] isolates the ad library in a sandbox that controls its permissions and file accesses, by hooking into any sensitive API calls made by those libraries. This method is effective even with code obfuscation, Java reflection and dynamic code execution. Developers can insert AdCapsule by adding a few lines to the source code of apps, and define security policies for ad libraries. This technique can block any unnecessary API calls made by ad libraries, such as reading files from the phone gallery. NoMoAds [180] takes a very different approach. It monitors network traffic on the device, and with Deep Packet Inspection it searches for traffic features typically found in ad libraries. When certificate pinning is used, the TLS connection cannot be decrypted but NoMoAds can still analyze information from the TCP/IP packet headers. When traffic is flagged as advertising, the connection is blocked.

5.8. Third Parties

The app platform and the OS may be taken as trusted entities trying to protect the user data they handle, against external actors. This is precisely what we address in this category, as we capture techniques to protect the security of the data when it is at rest in the device, in transit through the network, or at the server. We also capture techniques to protect user’s privacy in the presence of bystanders or nearby cameras that could visually reach the screen contents.

Device data. For the protection of the data at rest against local attackers (attack *H3*), mobile devices provide encryption for local storage [112], [181], and for local backups in iOS [182]. For Apple, although device encryption was introduced in iPhone OS 3, it was not until iOS 8 that the encryption key was bound to the user password, providing strong security [183], [184], [185]. For Android, full-disk encryption was introduced in version 4.4 but it has been replaced by file-based encryption. Given that dating apps typically store information locally with an unencrypted database at the application level [186], [187], OS-level encryption becomes more relevant. However, its security depends on the strength of the chosen password. We note that even when using strong keys, a device that is stolen while it is unlocked exposes private information. Storage encryption might also be bypassed by exploiting vulnerabilities [188].

Ephemeral messaging, which was covered in §5.2, is not only helpful in potentially malicious recipients, but also when a device containing sent/received nudes is accessed by third parties long after the exchange of the material (attacks *H1*, *H3*, *H4*). Besides, devices can be erased remotely with the Find My Device feature [189], [190], even for switched-off devices under certain conditions [191], [192], which helps protect the information in case of a stolen, lost or seized phone (attacks *H1*, *H3*). Google plans to extend this feature on Pixel devices in a way that the remote wipe can be triggered even if the device is powered off. This might be accomplished by keeping the Bluetooth chip active at all times [191]. Other strategies include biometric or PIN authentication for access to certain pictures (attacks *H1*, *H2*, *H4*), or placing the pictures in a less accessible album (attack *H4*). For instance, authentication may be required for opening the app or for accessing in-app saved pictures [193], [194]. The system photo gallery app may feature hidden albums [195] and albums that do not appear in the main panel [196].

Transport security. App data in transit containing nudes may be eavesdropped, modified, eliminated, re-ordered, replayed, or delayed by authorities, carriers, rogue hotspots, and other parties. TLS is the de facto standard protocol to encrypt and authenticate communications between user and server (attack *I1*), being HTTPS (HTTP over TLS) the secure protocol for web connections. Although there are ongoing efforts to migrate from TLS1.2 to TLS1.3, we will not analyze that aspect as it is not motivated by severe security issues [197].

Patsakis et al. [198] found in 2015 that in 14 out of the 18 surveyed iOS dating apps, photos were exchanged over plain HTTP, albeit now Apple requires TLS by default, except justified exceptions [199]. Hayes and Snow [32] found in 2018 that the Grindr iOS app was making plain HTTP requests to external domains associated with advertising and monetization, although the information exchanged with those parties did not contain images. Furthermore, Razaghpanah et al. [200] performed a large-scale study of 7,258 Android apps in 2017, in which they found that 50% of Android apps mix HTTP and HTTPS requests. They observed that Viber and Wire (which are within our surveyed apps) do not use forward-secret key exchange algorithms. Wire later added forward secrecy support to HTTPS [201]. We must note that since those app surveys are relatively old, it is possible that those issues have been resolved, either partially or completely.

To mitigate attacks against Certification Authorities (attack *I2*), such as in the 2011 Comodo cyber-attack [202], certificate pinning is a security mechanism that works as a replacement or addition to trusting PKI root certificates, where apps include, either hardcoded in the binary or in a local configuration file, a custom trusted certificate. The app checks that one of the certificates in the remote chain matches the pinned certificate. A survey of 5,079 unique apps from the official app stores found that only 2% and 6% of popular social network apps use certificate pinning, in Android and iOS respectively [203]. Shetty et al. [30]

collected evidence that Tinder, Happn, and Badoo do not use certificate pinning.

TLS, certificate pinning, and E2EE are especially relevant to protect the confidentiality of chats in countries where homosexuality, adultery, or other sexual activities are prosecuted. It has been reported that Egyptian authorities use fake or seized accounts on Grindr to trap gay people [204], although we are not aware of any PiTM attack by a nation-state actor with such purpose. For privacy and anti-censorship, and given that HTTPS does not hide the domain name⁶ and IP address, users can resort to TOR or anonymity VPNs, though not always supported by apps [205], [206].

Server data. We observe that all approaches we found useful to combat semi-honest apps in §5.7 are also effective for vulnerable apps in the face of third party adversaries (since they can potentially hijack the app). However, here we will focus only on vulnerable but honest software.

Android and iOS let users synchronize photos and store backups to a personal cloud space [207], [208]. This entails risks, as when nudes of female celebrities were exposed in the 2014 iCloud cyber-attack [209]. To mitigate server data breaches (attack J1), Android has end-to-end encrypted backups, and specific folders can be excluded from the backup process [210]. Specifically, starting from version 9.0, Android’s automatic device backups on the Google Drive are end-to-end encrypted, and the device’s PIN, pattern, or password is used to derive the encryption key [210]. Besides that, from the phone UI, Android lets users select which folders to back up, although there is no possibility of excluding specific images from the synchronization process. It additionally lets developers to control which files are included in the backup process of the app data, using XML rules or a BackupAgent [210]. In iOS, both backups and synchronized photos can be end-to-end encrypted [211], [212], except in the UK due to recent political developments [213]. Specifically, Apple’s more recent Advanced Data Protection feature lets users activate encryption of iCloud data belonging a number of categories, including the Photos app and the Device Backup, with E2EE [211], [212]. With the previous Standard Encryption, the encryption keys were stored in Apple-controlled HSMs (Hardware Security Modules). As an additional layer of protection, two-factor authentication can be activated for access to cloud data (attack J2).

Regarding the app’s server data persistence, several academic proposals allow to increase the guarantees that the data is self-destroyed after some average time [214], [215], [216], [217], usually around the idea of splitting the encryption key with secret sharing and distributing the shares in a cluster called DHT (Distributed Hash Table). The DHT will probabilistically overwrite the key after some time. It will do so in a natural way, because storage indices are allocated pseudo-randomly at each node’s memory, and the allocated memory is fixed and very limited.

Visualisation privacy. It is desirable to prevent surrounding third parties from spying (attack K1), catching

sight (incident K2), or externally capturing (attack K3), either intentionally or unintentionally, a nude displayed on the screen. This can occur with nearby CCTV cameras, shoulder surfing, and screen-cam. We consider that any mechanism that claims to provide visualisation privacy has to make the information more available (or only available) to the intended user. In other words, it must add some visualisation advantage for the intended user. Sender’s privacy is at stake when selecting the picture and when it is displayed in the chat history, as well as when the recipient opens it. Recipient’s privacy is affected if others oversee that is looking at other people’s nudes or that is using a dating app.

Hardware-based approaches have been developed to enhance screen privacy. This includes solutions to reduce the angle of vision, ranging from off-the-shelf screen protectors to a theoretical model for a two-layer auto-multiscopic display (these are the displays originally meant to represent three-dimensional scenes) [218], projecting different images in each direction, in which the original image can only be seen from an appropriate angle relative to the screen surface. In the realm of wearables, we have the smart glasses and the extended-reality headsets, but because of their high price, adoption is becoming slow.

Software-based strategies against incident H4 in Device Data also apply in K1,K2 because, for instance, placing pictures in a hidden album prevents glances by others. Telegram has the Hidden Media feature, in which the sender requires the recipient to tap the image to unblur it: “*Now you can also cover photos and videos with a shimmering layer that blurs the image. It will take your recipients one tap to break the spell and see the content*” [45]. For even more privacy, Confide’s Confidential Mode makes the user press and hold the touch screen to view received images [153]. However, this approach relies on the cooperation of the sender user, in the sense that the feature has to be enabled on a per message basis. The tap-to-open strategy mentioned in §5.6 for reception control has a similar effect as well. However, we consider reception control to be a dominant factor because we could assume that a consenting recipient generally uses the phone alone or in a private environment, but we should not assume that a recipient who is alone is a consenting recipient.

In Chen et al.’s academic proposal [219], bystanders only see a single-color area when the screen is viewed outside of the designed range. Its HideScreen system uses grid patterns to neutralize the low-frequency components of the image. The algorithm has been tested on mobile devices but only with grayscale images.

Alternatively, as a proactive strategy, one can use the own phone camera to detect bystanders, based on gaze estimation and measures of the distance between eyes [220], [221], [222], to then take appropriate action such as alerts or screen pre-processing. iScreen [220] is an academic proposal that actively tries to detect the presence of bystanders. It uses the phone’s camera to calculate the distance of bystanders by measuring the distance between eyes. With that information plus the ambient light sensor, it can adjust the color scheme

6. Except with secure DNS plus special extensions in TLSv1.3.

and brightness to reduce shoulder surfer’s vision. Saad et al. [221] suggest to apply existing algorithms on the gaze so as to evaluate if an additional face detected by the phone camera is actually looking at the screen. PrivacyScout [222] suggests to further reduce the alert fatigue caused by false positives, with a risk assessment score built from a trained model that uses face size, distance between eyes, and gaze angle, among other variables. This method has been tested with an external camera. It is designed for phone cameras equipped with wide angle lenses.

Other proposals enable to detect devices in close proximity with Bluetooth and/or WiFi RSSI (Received Signal Strength Indicator) measurements. Devices closer than two meters can be detected [223], and, with environment calibration, even foot-level accuracy over short distances [224]. BlueEye [224] is a system to detect nearby devices using Bluetooth. Although their proposal is geared towards the measurement of crowdedness at a particular location, it might be used to prevent screen-cam. In [223], the authors present a machine learning approach to detect devices that are closer than two meters, based on WiFi and Bluetooth fingerprints. Proximity detection mechanisms can of course be combined with screen rendering algorithms, so that when a bystander is detected the screen switches to privacy mode. Other works [225], [226] implement alert methods that just notify the user about the presence of bystanders, by displaying a screen icon, a message, or a preview of the shoulder surfer.

We finally observe that, for the screen-cam scenario, the recapture detection algorithms we discussed in Forwarding and Posting Policies in §5.2 work equally for malicious recipients than for other malicious third parties.

5.9. Offline Tools

Educational apps, parental features, support apps/sites, and in-app guidelines play an important role towards safer sexting. Such tools work ‘offline’, in the sense that they are not necessarily tied to a real and specific sexting activity.

Educational apps. Sexting and privacy education activities can be supported by special-purpose apps [227], [228], [229]. Patchin and Hinduja [230] emphasize that there should be a sexting education as much as there should be a sex education. Sexting education is not about encouraging (or discouraging) teenagers to engage in sexting. It simply acknowledges the reality that they are sexually curious and that they use technology for that.

Talk About Sex [227] is a multiplayer mobile game app for adolescents and youth workers in sexual education workshops. The game prompts for informal tasks in turns, for example: “Take a selfie on someone else’s phone”, or “Take a photo of a body part”. It has been tested with groups of adolescents from 13 to 19 years old. The permissive approach of the app successfully managed to engage the participants, especially the younger ones, and broke the ice for discussing about sexuality topics.

iBuddy [228] is a mobile game tested with young people from 16 to 20 years old that is to be used in school

education activities conducted by a presenter. Once launched by the participants, the app claims that a new ‘virtual buddy’ will be incorporated into the class. In the background, the app temporarily collects private data, including phone gallery pictures previously filtered by AI algorithms, to a local server. The app creates a wow-effect by prompting questions to the participants that imply that their personal data has been collected. For example, if the participant has cat pictures, it prompts, “[participant name] seems to like pets. Will your new buddy also share this passion?” This activity makes participants experience privacy issues themselves during the game.

FoxIT [229] is an Android app tested on adults from 18 to 54 years old. It consists of two modules. The first module analyzes and reports the enabled permissions (especially the ones related to sensitive information) for the apps installed on the participant’s phone, giving a link to the corresponding configuration panel of Android permissions, for subsequent review. The second module is an animated set of privacy lessons around several topics, such as encrypted messaging or messaging apps.

Parental supervision. In social networks, there are parental guides and tools for the supervision and education of children, which can include the linking of a minor’s account with the parent’s or guardian’s account [231], [232]. Parents can then monitor the list of followers, get notified if their teen reports someone (if the teen chooses to share that information), and other controls [233]. Furthermore, parents may have the ability to approve or deny, at their discretion, requests to change privacy settings in supervised accounts, such as switching the account from private to public, loosening restrictions for sexual content, or the ability to receive DMs from people the teen is not following or connected to [234].

Support apps and sites. There are platforms where teens can get personal support for mental health, suicide prevention, or sexting-related issues. For instance, in the Teen Talk app [235], teens aged 13-19 can get advice anonymously from trained teen advisors.

Safety guidelines. Safety guidelines of dating apps provide tips to identify fake profiles, advice about physical safety in in-person meet-ups and in countries where homosexuality is prosecuted, and information about digital abuse.

6. Limitations

To narrow the technical scope, in our app review we skipped web-only platforms, such as OnlyFans and escort sites, despite being fairly standard use cases. Nevertheless, the web space is a much more decentralised platform than the mobile apps, which would make it more challenging to assess what are the top messaging, dating, escort, and content-creator websites. Also for the sake of simplicity, our study scenario overlooked desktop and laptop computers, multiparty conversations, multiparty pictures, video, and sharing over cloud drives.

The chosen search strings, the specific harms, threats, attacks and incidents we considered as in-scope, and the criteria to derive our taxonomy, as well as our judgment

on the applicability of every academic proposal, can be object of further revision. For instance, we did not cover: viruses spread over sexts, recipients showing their screen to others, malicious moderators leaking the content they inspect, and concerns of being depicted in sexual deepfakes shared among other parties.

Finally, the definition of nudity used, the language chosen for search queries, and the configured app store country were conditioned by the authors’ background, possibly introducing a western (W.E.I.R.D.) bias to the scope of results.

7. Discussion

In this section we discuss the main take-aways from our study, and provide directions for future research and industry development in this important topic.

7.1. Ephemeral messaging

Prior research indicates that Snapchat is the most used app in sexting [24], [33]. It was launched back in 2011 with its disappearing ‘snaps’ [236], which can be considered the birth of ephemeral messaging. A 2020 user survey reported ‘disappearing messages’ as the most popular messaging feature in sexting [24]. In our app survey, half of the analyzed apps implement some ephemeral messaging feature (we identified 44 feature instances in total), ranging from apps that support three kinds of ephemeral messaging features to apps that support none. In 2022, Grindr reported a 72% increase in expiring photos sent [237]. It also reported the sharing of 1 billion albums in 2023, and 2+ billion in 2024 [238]. We note that academic proposals and app features on ephemeral messaging jointly cover three categories in our threat modeling, namely: Post Exposure User-to-User Privacy, Third Parties, and even App Platform when controlled by a middleware instead of the app itself. Such observations help explain and confirm the popularity of ephemeral messaging and its close ties to sexting.

❓ *Ephemeral messaging and other privacy features are valuable even if they are not bullet-proof against savvy hackers or if they assume cooperative behavior by a majority of users. However, since it is an essential safer sexting feature, and since the vast majority of apps are vulnerable to interception and tampering attacks, certificate pinning and existing app integrity APIs can reinforce its security.*

7.2. End-to-End Encryption

Besides ephemeral messaging and image information reduction, E2EE is the other technique that covers three different threats within our model. In the case of E2EE, it covers: *G* (e.g., the app could covertly exploit the images shared among users to train AI models for nude detection), *I* (e.g., a nation-state adversary could obtain in-transit data of dating apps containing nudes), and *J* (e.g., an attacker could target a vulnerable server to access nudes, as with the 2014 iCloud cyber-attack). For *G* and *I*, it is more difficult

to identify real-word instances of attacks, possibly because they can be executed leaving a minimal or no trace. In all three threats, a single attacker can target a relatively large amount of users over a short time span.

❓ *Such observations are consistent with the long-held view that E2EE is a fundamental security and privacy feature. For compatibility, features that need to operate on image contents (e.g., nude detection for reception control) will have to be executed on-device or on encrypted domain.*

7.3. Layering in threat C

Threat *C* (recipients spreading information to others or threatening to) is a central issue in sexting, since it affects multiple harm scenarios (doxxing, sextortion, cyberbullying, revenge porn, exploitation of the material), and it can lead to serious consequences such as suicide or anxiety. It is actually the most studied threat in the Related Work of §3. We observe that attack strategies can be escalated on top of others: using a profile with false information, then putting pressure to share nudes (*A1*), then using hacking techniques to retain data against sender’s will (*B3*), then threatening to spread information, and finally disclosing the nudes to others (*C1*). Correspondingly, our taxonomy suggests a multi-layered strategy: sexting education and parental supervision in Offline Tools, image privacy features/strategies and reliable profile information in Pre Exposure, integrity attestation in Adversarial-Recipient Security, ownership verification and ephemeral messaging in Post Exposure, and user reporting and forensic digital watermarking in Accountability.

❓ *Since no single technique appears to be the ultimate solution, effectively fighting against NDII and related threats requires a multi-layered strategy, from sexting education to post-facto accountability techniques such as digital watermarking. The techniques are extraordinarily diverse, showing that safer sexting goes far beyond nude detection.*

7.4. Usability difficulties

In E2EE, reception control, and ephemeral messaging, we have identified examples where inconsistencies in the labeling (in the UI) and the naming (in the documentation) of features may lead to confusion to users, that is: discrepancies between the labeling and the naming within an app, a feature being labeled the same in two apps but with significantly different specifications, or the same underlying function being labeled differently in two apps. There can also be platform-specific marketing policies (a feature may be free of charge in some apps while premium in others), and non-uniform feature combinations across apps (as a trade-off between privacy and usability, some apps apply UI restrictions only to ephemeral messages). Strategies for Reception Control features are diverse, e.g., filtering, tap-to-open, blurring, NSFW, or first contact preferences. For Offline Tools, Instagram’s release log for teen support and parental supervision [239], to give

an example, shows a constantly changing landscape of features, which can be overwhelming for users to navigate through. We finally add that the median year of ephemeral feature roll-out (within our surveyed apps) is 2020.5.

💡 *The diverse naming, labeling, specifications, marketing policies, and combinations of features, added to the fast pace of new releases, poses usability difficulties. Apps can make use of wizards, explanatory videos, in-app contextual help, or packaging of multiple features to better assist users. Prior research [25] suggests the use of safety settings preset configurations for nudes, to avoid having to configure multiple privacy settings for every piece of content.*

7.5. Privacy preservation

For the user to be visible by and see only the people deemed trustworthy and of interest, social networks and dating apps may have the friends in common, user-granted badges, date joined, and other profile data. If the truthfulness of the information (such as age, marital status, or country of residence) is to be certified by a third-party entity, there is the privacy challenge of preventing that the app gains extra knowledge, and of preventing the certifying entity from knowing the fact that some dating app is being used or the connection times. If dating platforms interact with social networks via API to let users filter co-workers and relatives, or to verify the number of friends in common in the social network, there is the challenge of preventing those platforms from mutually revealing any extra information.

Systems based on digital watermarking, digital signatures, perceptual hashing and/or blockchain can be used to prove ownership in requests to take pictures down from websites, check ownership before allowing to share over a platform, or determine the origin of a leak after sharing with multiple recipients. This would benefit recreational sexters and sex workers alike. If such systems use a trusted third party, this adds privacy concerns. Initiatives such as `stopncii.org` (which uses on-device PDQ algorithm for perceptual hashing) and `takeitdown.ncmec.org` (of which we do not know the algorithm in use) are a step forward but still face usability issues or lack of transparency for external analysis. Also, E2EE chats are outside of their scope.

💡 *For systems using perceptual hashing databases, external social networks, certification entities of citizens' demographic data, smart contracts, or trusted third parties, preserving privacy is a key challenge, in the sense of not mutually leaking any extra information. Other issues may relate to performance, usability, and transparency.*

7.6. Applicability analysis

From within our collection of 138 selected academic and gray literature, set aside nude detection algorithms, 12 mention sexting- or dating-related scenarios as a use case. Besides, 19 other papers mention private messaging as a use case or a concern, half of which address server

data protection. Digital watermarking has a large literature corpus but, historically, the vast majority of considered use cases deal with copyright protection rather than sexting issues, possibly because digital camcorders and DVD players became popular before social networks. Similarly, image-encryption approaches are largely motivated by the use case of public or semi-public posts in social networks. There is also ongoing work about the usage of blockchain for image security and privacy in several projects, including within the JPEG Committee, although ongoing projects focus more on the media consumption chain and the creative sector [240]. In our collected user guides and feature documentation of social networks and messaging platforms, sexting- or dating-related scenarios are only explicitly mentioned in the nude detection and in the reporting of users.

💡 *There exist a vast number of applicable approaches, both within academia and the industry, but it appears that the specific use case of sexting has received limited attention. Our study can be used to analyze the applicability of proposals to our threat model or to frame them within our taxonomy, highlighting their value in the sexting context.*

8. Conclusion

In our breadth-driven, user-centric review, we strived to gather up-to-date information, merging academic proposals together with existing app features. From there, grouping the approaches by threat model and privacy properties (rather than by technology or user segment) produced an illustrative overview of the state of the art. Since the sharing of nudes is a very—if not the most—challenging scenario of pictures exchanged in a chat, it constitutes a solid foundation for constructing a taxonomy of methods to protect the exchange of media in chats. Our study found a fragmented ecosystem of app features and an extensive variety of techniques, none of which representing the ultimate solution, but each contributing to a safer sexting in a different way.

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Appendix A. Search strings

For our literature review we based our search strings on keywords that reflect various aspects of our RQ1 (and to a lesser extent, RQ3 and RQ4). We show below some of the keywords we used as building blocks of our search strings, covering seven different aspects:

- Necessary elements: The nude image artifact. Note that we expanded synonyms and core concepts, e.g. *photo OR image OR picture*.
- Population: It can be general (*users*) or communities (e.g., *sex workers, women, religious*).
- Intervention: Interventions can be of very diverse nature, as in *certificate pinning, image encryption*, and so on.
- Comparison: *survey, SoK, literature review*.
- Harms and attack scenarios: Examples include *sexortion, cyberbullying, sexual deepfakes*.
- Outcomes: *security OR privacy OR accountability*, and in some cases we used *protection*.
- Context: *dating, mobile apps, private chat, sexting*.

We crafted a list of generic search strings covering only two or three different aspects above (SS-G), plus a list of more specific search strings about high-level speculated mechanisms (SS-S), about sexting-related harms or attacks (SS-H), and some others more oriented to at-risk or at-concern communities (SS-C), resulting in the final list below. Search strings are relatively short and almost always contain one () clause at most, so as to keep them fairly balanced in terms of length and semantic load.

SS-G01 images mobile apps (privacy OR security OR accountability)
 SS-G02 pictures mobile apps (privacy OR security OR accountability)
 SS-G03 photos mobile apps (privacy OR security OR accountability)
 SS-G04 "auto pornography" (privacy OR security OR accountability)
 SS-G05 sexting (privacy OR security OR accountability)
 SS-G06 nudes (privacy OR security OR accountability)
 SS-G07 dating apps (privacy OR security OR accountability)
 SS-G08 adversarial dating (apps OR users)
 SS-G09 messaging apps (photo OR image OR picture) privacy
 SS-G10 dating apps (photo OR image OR picture) privacy
 SS-G11 social network (photo OR image OR picture) privacy
 SS-G12 images ios mobile apps
 SS-G13 images android mobile apps
 SS-G14 privacy images mobile apps
 SS-G15 image privacy without authentication
 SS-G16 dating apps auto pornography privacy
 SS-G17 dating apps image privacy
 SS-G18 dating apps sexting privacy
 SS-G19 dating apps security
 SS-G20 dating apps auto pornography security
 SS-G21 dating apps image security
 SS-G22 dating apps sexting security
 SS-G23 adversarial dating apps
 SS-G24 image privacy survey papers
 SS-G25 image privacy SoK papers
 SS-G26 privacy photo sharing
 SS-G27 literature review survey image privacy mobile apps
 SS-G28 (image OR picture) private chat
 SS-G29 private (image OR picture) chat message
 SS-G30 sexting survey apps
 SS-G31 autopornography (privacy OR security OR accountability)
 SS-S01 transport security dating apps
 SS-S02 https dating apps
 SS-S03 certificate pinning dating apps
 SS-S04 transport security https certificate pinning
 SS-S05 end-to-end encryption dating apps

SS-S06 mobile apps third party (threats OR security OR privacy)
 SS-S07 mobile apps sponsor (threats OR security OR privacy)
 SS-S08 mobile apps monetization (threats OR security OR privacy)
 SS-S09 cloud data protection apps
 SS-S10 cloud data dating apps
 SS-S11 device data protection apps
 SS-S12 local storage dating apps
 SS-S13 mobile privacy (screen OR eyewear)
 SS-S14 hidden media privacy blur
 SS-S15 confirmation opening image
 SS-S16 (pornography OR nude) detection encrypted images
 SS-S17 consent unsolicited dick pics
 SS-S18 (pornography OR nude OR sexual content OR skin tone OR explicit content) detection
 SS-S19 mobile screen privacy
 SS-S20 shoulder surfing privacy
 SS-S21 screen-cam (image OR photo OR picture) privacy
 SS-S22 mobile phone proximity detection
 SS-S23 detect camera (printout OR screen)
 SS-S24 camera detect liveness
 SS-S25 image recapture detection
 SS-S26 educational tools (photo OR image OR picture) privacy
 SS-S27 sexting education mobile app
 SS-S28 dating apps safety guidelines sexting
 SS-S29 blur face image privacy
 SS-S30 social networks number of friends in common
 SS-S31 dating apps verified profile
 SS-S32 dating apps fake profiles
 SS-S33 TEE image (privacy OR security)
 SS-S34 DRM image (privacy OR security)
 SS-S35 play integrity app attest image privacy
 SS-S36 "SafetyNet" app attest image privacy
 SS-S37 per-image app permissions
 SS-S38 (phone OR image) gallery permissions mobile phone
 SS-S39 app photo library access
 SS-S40 app image gallery access
 SS-S41 (image OR photo OR picture OR JPEG) encryption
 SS-S42 mobile app user interface restrictions (images OR screen)
 SS-S43 mobile app restrict saving content
 SS-S44 (image OR photo OR picture) privacy screenshots
 SS-S45 privacy (image OR photo OR picture) intents
 SS-S46 (image OR photo OR picture) ownership verification mobile app forwarding
 SS-S47 (photo OR image OR picture) (authenticity OR authentication)
 SS-S48 (photo OR image OR picture) manipulation detection
 SS-S49 artificial intelligence generated image detection
 SS-S50 ephemeral (images OR pictures OR photos OR media) privacy mobile app
 SS-S51 self-destructing (images OR messages)
 SS-S52 disappearing (images OR messages)
 SS-S53 time-expiring (images OR messages)
 SS-S54 burn-after-reading (images OR messages)
 SS-S55 unsend (images OR messages)
 SS-S56 "delete for everyone"
 SS-S57 private album dating apps
 SS-S58 screen-cam robust digital watermarking
 SS-S59 protective watermark
 SS-S60 blockchain (image OR photo OR picture) privacy
 SS-S61 screenshot alerts mobile app
 SS-S62 sex workers clients vetting
 SS-S63 strip metadata (image OR photo OR picture OR JPEG)
 SS-H01 cyberbullying sexting technology (prevention OR protection OR accountability)
 SS-H02 sexual deepfakes technology (prevention OR protection OR accountability)
 SS-H03 sextortion technology (prevention OR protection OR accountability)
 SS-H04 revenge porn technology (prevention OR protection OR accountability)
 SS-H05 catfish sexting technology (prevention OR protection OR accountability)
 SS-H06 non-consensual sexting technology (prevention OR protection OR accountability)
 SS-H07 cyberflashing technology (prevention OR protection OR accountability)
 SS-H08 doxxing technology (prevention OR protection OR accountability)
 SS-H09 pornography (exploitation OR trafficking) technology (prevention OR protection OR accountability)
 SS-H10 cyber grooming sexting technology (prevention OR protection OR accountability)
 SS-C01 (children OR teenagers OR adolescents) technology sexting privacy
 SS-C02 (LGBTQ+ or "sexual minorities") technology sexting privacy
 SS-C03 (politicians OR celebrities) technology sexting privacy
 SS-C04 "intellectually disabled" technology sexting privacy
 SS-C05 "sex workers" technology sexting privacy
 SS-C06 women technology sexting privacy
 SS-C07 (religion OR faith) technology sexting privacy
 SS-C08 (black OR hispanic OR ethnicity OR race) technology sexting privacy
 SS-C09 "young adults" technology sexting privacy

Appendix B.

Academic and gray literature presenting or explaining some approach

- #1 "It's stressful having all these phones": Investigating sex workers' safety goals, risks, and practices online
- #2 "Disadvantaged in the American-dominated Internet": Sex, work, and technology
- #3 A bag-of-features approach based on hue-sift descriptor for nude detection
- #4 A blind watermarking method using maximum wavelet coefficient quantization
- #5 A compression sensing and noise-tolerant image encryption scheme based on chaotic maps and orthogonal matrices
- #6 A decentralised messaging system robust against the unauthorised forwarding of private content
- #7 A forensic analysis of images on online social networks
- #8 A holistic approach to undesired content detection in the real world
- #9 A hybrid robust image watermarking method based on DWT-DCT and SIFT for copyright protection
- #10 A new blockchain-based trusted DRM scheme for built-in content protection
- #11 A new color image encryption scheme using CML and a fractional-order chaotic system
- #12 A new JPEG encryption scheme using adaptive block size
- #13 A new wavelet based logo-watermarking scheme
- #14 A novel image encryption scheme based-on JPEG encoding
- #15 A peer-to-peer agent community for digital oblivion in online social networks
- #16 A perceptually scalable and JPEG compression tolerant image encryption scheme
- #17 A robust content based digital signature for image authentication
- #18 A robust DCT-based watermarking for copyright protection
- #19 A secure data self-destructing scheme in cloud computing
- #20 A secure document self-destruction scheme with Identity Based Encryption
- #21 A secure document self-destruction scheme: An ABE approach
- #22 A secure self-destructing scheme for electronic data
- #23 A self-destructing secure messaging system using multi key management scheme
- #24 A semifragile watermark scheme for image authentication
- #25 A two-layers automultiscopic display as a resistant device to shoulder-surfing attacks
- #26 A watermarking protocol based on Blockchain
- #27 AdCapsule: Practical confinement of advertisements in Android applications
- #28 AI vs. AI: Can AI detect AI-generated images?
- #29 An adaptive digital image watermarking technique for copyright protection
- #30 An efficient retrograde storage for self-destructing messages on frequently colliding hash table
- #31 An efficient watermarking method based on significant difference of wavelet coefficient quantization
- #32 An experimental tattoo de-identification system for privacy protection in still images
- #33 An improved stegastamp watermarking scheme by screen-cam robust detector
- #34 An improved SVD-based watermarking scheme for protecting rightful ownership
- #35 An improved SVD-based watermarking technique for copyright protection
- #36 An SVD-based watermarking scheme for protecting rightful ownership
- #37 Anti-screenshot watermarking algorithm for archival image based on deep learning model
- #38 Are they actually looking? Identifying smartphones shoulder surfing through gaze estimation
- #39 Automatic localization and detection applicable to robust image watermarking resisting against camera shooting
- #40 Bitstream-based JPEG image encryption with file-size preserving
- #41 Blind dual watermarking for color images' authentication and copyright protection
- #42 BlueEye: A system for proximity detection using Bluetooth on mobile phones
- #43 Bootstrapping trust in online dating: Social verification of online dating profiles
- #44 Can messaging applications prevent sexting abuse? A technology analysis
- #45 Classification of captured and recaptured images to detect photograph spoofing
- #46 Cocktail watermarking for digital image protection
- #47 Collaborative privacy management: Mobile privacy beyond your own devices
- #48 Communicating shoulder surfing attacks to users
- #49 Constrained R-CNN: A general image manipulation detection model
- #50 Cryptagram: Photo privacy for online social media
- #51 Cryptoeyes: Privacy preserving classification over encrypted images
- #52 DatingSec: Detecting malicious accounts in dating apps using a content-based attention network
- #53 DeepScan: Exploiting deep learning for malicious account detection in location-based social networks
- #54 Design and implementation of image encryption algorithm by using block based symmetric transformation algorithm (hyper image encryption algorithm)
- #55 Digital image authentication from JPEG headers
- #56 Digital watermarking technique for protecting digital images
- #57 Efficient and syntax-compliant JPEG 2000 encryption preserving original fine granularity of scalability
- #58 Efficient selective encryption for JPEG 2000 images using private initial table
- #59 Encrypted domain skin tone detection for pornographic image filtering
- #60 End-to-end encrypted messaging protocols: An overview
- #61 Epson Image Authentication System, IAS
- #62 Explicit content detection system: An approach towards a safe and ethical environment
- #63 Explicit content detection using faster R-CNN and SSD MobileNet v2
- #64 Explicit content image detection
- #65 Explicit image detection using YCbCr space color model as skin detection
- #66 Fake image detection using machine learning
- #67 Fast frequency domain screen-shooting watermarking algorithm based on ORB feature points
- #68 Fides: Unleashing the full potential of remote attestation
- #69 File system design with assured delete
- #70 Fourier image watermarking: Print-cam application
- #71 FoxIt: Enhancing mobile users' privacy behavior by increasing knowledge and awareness
- #72 Go-sharing: A blockchain-based privacy-preserving framework for cross-social network photo sharing
- #73 Image authentication by assessing manipulations using illumination
- #74 Image authentication by detecting traces of demosaicing

#75 Image recapture detection using multiple features
 #76 Image recapture detection with convolutional and recurrent neural networks
 #77 Image recapture prevention using secure display schemes on polarized 3D system
 #78 iScreen: A pure software-based screen privacy protection system for mobile devices
 #79 Joint image encryption and screen-cam robust two watermarking scheme
 #80 JPEG encryption for image rescaling in the encrypted domain
 #81 JPEG encryption with file size preservation
 #82 JPEG fake media: A provenance-based sustainable approach to secure and trustworthy media annotation
 #83 JPEG image encryption with adaptive DC coefficient prediction and RS pairpermutation
 #84 JPEG image encryption with improved format compatibility and file size preservation
 #85 Keep others from peeking at your mobile device screen!
 #86 Learning rich features for image manipulation detection
 #87 Leveraging deep neural networks to fight child pornography in the age of social media
 #88 Making distributed mobile applications safe: Enforcing user privacy policies on untrusted applications with secure application flow enforcement
 #89 Male sex workers' strategies to manage client-related risks of violence
 #90 Metadata-based image watermarking for copyright protection
 #91 Multipurpose watermarking for image authentication and protection
 #92 NoMoAds: Effective and efficient cross-app mobile ad-blocking
 #93 Online detection of AI-generated images
 #94 Owner-customer right protection mechanism using a watermarking scheme and a watermarking protocol
 #95 P3: Toward privacy-preserving photo sharing
 #96 Pandora messaging: An enhanced self-message-destructing secure instant messaging architecture for mobile devices
 #97 Playing for privacy awareness: Learning from a "wow-moment" with iBuddy
 #98 Preserving privacy for Blockchain-driven image watermarking using fully homomorphic encryption
 #99 Print-cam robust image watermarking based on hybrid domain
 #100 Privacy-preserving photo sharing based on a public key infrastructure
 #101 PrivacyScout: Assessing vulnerability to shoulder surfing on mobile devices
 #102 Protecting against screenshots: An image processing approach
 #103 Protecting mobile users from visual privacy attacks
 #104 Puppies: Transformation-supported personalized privacy preserving partial image sharing
 #105 Risk, resilience and reward: Impacts of shifting to digital sex work
 #106 Robust privacy-preserving image sharing over online social networks (OSNs)
 #107 Robust watermarking of screen-photography based on JND
 #108 Rushmore: Securely displaying static and animated images using TrustZone
 #109 Safer digital intimacy for sex workers and beyond: A technical research agenda
 #110 SafeVanish: An improved data self-destruction for protecting data privacy
 #111 Screen-cam robust image watermarking with feature-based synchronization
 #112 Secure JPEG scrambling enabling privacy in photo sharing
 #113 Secure photo sharing in social networks
 #114 Securing image metadata using Advanced Encryption Standard
 #115 Selective encryption of JPEG images with chaotic based novel S-box
 #116 Self-destructing instant messages by employing KAD DHT network
 #117 Semantic remote attestation — A virtual machine directed approach to trusted computing
 #118 Sex talk: Designing for sexual health with adolescents
 #119 Shareff: A sticky policy middleware for self-destructing messages in Android applications
 #120 Short paper: Chips: Content-based heuristics for improving photo privacy for smartphones
 #121 Single-view recaptured image detection based on physics-based features
 #122 Skin sheriff: A machine learning solution for detecting explicit images
 #123 Smartphone proximity detection using WiFi and BLE fingerprinting
 #124 Social-Guard: Detecting scammers in online dating
 #125 SPE: Security and privacy enhancement framework for mobile devices
 #126 Specification and implementation of metadata for secure image provenance information
 #127 Survey on skin tone detection using color spaces
 #128 Time controlled cloud environment with self destructing data system (SeDas) for data confidentiality
 #129 Transparent and perceptually enhanced JPEG image encryption
 #130 Understanding and integrating Kodak picture authentication cameras
 #131 User-driven access control: Rethinking permission granting in modern operating systems
 #132 Vanish: Increasing data privacy with self-destructing data
 #133 Watermark-preserving keypoint enhancement for screen-shooting resilient watermarking
 #134 Watermarking digital images for copyright protection
 #135 Watermarking for image authentication
 #136 Watermarking for intellectual property protection
 #137 X-pire! — A digital expiration date for images in social networks
 #138 You are how you click: Clickstream analysis for sybil detection

Appendix C.

Surveyed apps

	App	User-tested empirically?	Documentation checked?	Allows picture sharing?	Observations
#1	Badoo	Yes	Yes	Yes	
#2	BeReal	Yes	Yes	Yes	
#3	Boo	Yes	Yes	Yes	
#4	Bumble	Yes	Yes	Yes	
#5	ChristianMingle	Yes	Yes	No	
#6	Confide	Yes	Yes	Yes	
#7	Daddyhunt	Yes	Yes	Yes	
#8	Dinder Club	No	Yes	No	Only persons with intellectual disabilities can join after a manual screening process.
#9	Discord	Yes	Yes	Yes	
#10	Facebook*	No	No	N/A	Uses Facebook Messenger for the messaging.
#11	Facebook Lite*	No	No	N/A	Uses Facebook Messenger for the messaging.
#12	Facebook Messenger	Yes	Yes	Yes	
#13	Google Meet*	No	No	N/A	Only for video meetings.
#14	Grindr	Yes	Yes	Yes	
#15	Happn	Yes	Yes	No	
#16	HER	Yes	Yes	Yes	
#17	Hinge	Yes	Yes	No	
#18	Hornet	Yes	Yes	Yes	
#19	Instagram	Yes	Yes	Yes	
#20	Jack'd	Yes	Yes	Yes	
#21	JSwipe	Yes	Yes	Yes	
#22	Kaboom*	No	No	N/A	Not available any more. Special case; not a messaging app per se.
#23	Kik	Yes	Yes	Yes	
#24	LesPark*	No	No	Unknown	The platform bans the use of the app by men. Lacking user guides.
#25	Meetic	Yes	Yes	Yes	
#26	Muzz/Muzmatch	Yes	Yes	Yes	
#27	OkCupid	Yes	Yes	No	
#28	PinkCupid	Yes	Yes	No	
#29	Pinterest	Yes	Yes	Yes	
#30	Plenty Of Fish	Yes	Yes	Yes	
#31	Recon	Yes	Yes	Yes	
#32	Reddit	Yes	Yes	Yes	
#33	Romeo/Hunqz	Yes	Yes	Yes	
#34	Scruff	Yes	Yes	Yes	
#35	Signal	Yes	Yes	Yes	
#36	Skout	Yes	Yes	Yes	
#37	Skype	Yes	Yes	Yes	
#38	Snapchat	Yes	Yes	Yes	
#39	Surge	Yes	Yes	Yes	
#40	Telegram	Yes	Yes	Yes	
#41	Tami	Yes	Yes	Yes	
#42	TikTok	Yes	Yes	No	
#43	Tinder	Yes	Yes	No	
#44	Trans	Yes	Yes	Yes	
#45	Viber	Yes	Yes	Yes	
#46	WeChat	Yes	Yes	Yes	
#47	Whatsapp	Yes	Yes	Yes	
#48	Whatsapp Business	Yes	Yes	Yes	
#49	Wickr	Yes	Yes	Yes	
#50	Wire	Yes	Yes	Yes	
#51	X/Twitter	Yes	Yes	Yes	
#52	Zoe	Yes	Yes	Yes	

*The app was not analyzed, neither empirically nor documentation-wise.

Appendix D.

App survey procedure

- 1) We created an account and looked at all available options when editing the profile, as well as in the general settings section within the app.
- 2) We sent a picture over a private chat between two test user accounts. We did tap-and-hold of the picture on the selection phase, of the Send button, and of the picture in the chat history on both sender and recipient. We looked at any contextual menu options that appeared.
- 3) We looked up official user guides and technical documents where the procedure of sharing messages, images and albums within that app was explained, also checking other subsections.
- 4) We checked the links to any other relevant features within the documents we came across.

Appendix E.

Evaluated and annotated properties of the approaches

- Publication year. For app features and gray literature, we used the Wayback Machine from the Internet Archive (<https://web.archive.org>) and advanced filters in a web search engine to infer the date of the earliest publication version of the document, or of an equivalent document with the relevant information.
- What threats of Table 2 are tackled? (multiple choice). We selected all that apply even when the document does not address them as a motivating use case.
- What are the provided benefits? (multiple choice). We considered: privacy for the sender, privacy for the recipient, confidentiality, data authenticity, software integrity, and accountability. Every approach provides at least one benefit.
- When is the mechanism enforced? (single choice). Options: before delivering the image to the recipient, after delivering the image to the recipient, or it does not present a mechanism that intervenes directly in specific image exchanges.

Appendix F.

In-use tools, solutions and app features

NOTE: The links point to the corresponding technical reference or to an implementation example.

- #1 [Communication Safety \(Apple\) \[13-17yo sending nudes\]](#)
- #2 [SensitiveContentAnalysis API \(Apple\) \[sending nudes\]](#)
- #3 [Using a third-party app to manually view and remove metadata/EXIF information from pictures](#)
- #4 [Signal automatically removes metadata/EXIF from pictures \(not officially documented, though\)](#)
- #5 [In-app drawing tools, or system photo editing tool, or third party photo editing apps \(redacting picture for de-identification: cropping, masking, etc.\)](#)
- #6 [Account badge based on other users' inputs/feedback](#)
- #7 [Profile verification using automated facial recognition and/or humanity tests](#)
- #8 [Profile verification performed or assisted by server-side human moderators](#)
- #9 [Dinder Club \[in-person or manual screening to certify intellectual disability\]](#)
- #10 [Connecting Instagram to Tinder \[linking accounts across platforms\]](#)
- #11 [Continuing to Make Instagram Safer for the Youngest Members of Our Community \[mutual follow requirement in children-adults contacts\]](#)
- #12 [Continuing to Make Instagram Safer for the Youngest Members of Our Community \[notify children about "potentially suspicious behavior" of adult users, in DMs\]](#)
- #13 [Who you can send messages to on Instagram \[separate, text-only inbox for first-time messages sent from people who the user does not follow\]](#)
- #14 [Requirements for starting a chat, based on recipient's preferences \(age, location, etc.\) \[First Contact preferences\]](#)
- #15 [Mutual match requirement to be able to chat with someone, in swipe-based dating apps](#)
- #16 [Only women can start a conversation after a match, in swipe-based dating apps \[First Move feature\]](#)
- #17 [The app platform proactively purging the userbase from malicious profiles, especially scammers](#)
- #18 [Multiprofile support \(either via software or by having separate devices\) \[for sex workers to maintain completely separate online personas\]](#)
- #19 [Why do you also need my LinkedIn login? \(The League\) \[filter coworkers\]](#)
- #20 [Extended profile information such as number of followers, friends in common, list of friends, and date joined or member since.](#)
- #21 [Procedures \(such as providing a valid credit card\) and AI-powered algorithms aimed at verifying \$\geq 18\$ in dating apps and \$\geq 13\$ in social networks](#)
- #22 [Good-behavior badges granted by other users](#)
- #23 [National Ugly Mugs, UK \[potential client checking/vetting tool/service for sex workers\]](#)
- #24 [Bad Client and Aggressors List \[potential client checking/vetting tool/service for sex workers\]](#)
- #25 [Asking potential clients to share several selfies \[verification strategy used by sex workers\]](#)
- #26 [Screenshot/Recording prevention in ephemeral media, or in any exchanged media](#)
- #27 [Save and/or forward prevention in ephemeral media, or in any exchanged media](#)
- #28 [Allowing to save the received pictures within the app, but not to the phone gallery](#)
- #29 [Ephemeral media: Self-destructing time-expiring messages](#)
- #30 [Ephemeral media: Self-destructing view once pictures](#)
- #31 [Ephemeral media: User-controlled unsending](#)
- #32 [Ephemeral media: User-controlled albums](#)
- #33 [Play Integrity API \(Android\) \(Formerly, SafetyNet\)](#)
- #34 [App Attest API \(iOS\)](#)
- #35 [Perceptual hashing database to prevent the spread of nudes in posts and profiles \[takeitdown.ncmec.org, stopncii.org\]](#)
- #36 [Epson Image Authentication System, IAS \[digital signature from camera firmware\]](#)
- #37 [Understanding and Integrating KODAK Picture Authentication Cameras \[digital signature from camera firmware\]](#)
- #38 [Jack'd \[Sensitive Content Filter\]](#)
- #39 [Safer Messaging on Discord \[Sensitive Content Filter\]](#)
- #40 [With Bumble's Private Detector, You Have Control Over Unsolicited Nudes](#)
- #41 [NSFW preference in the user profile](#)
- #42 [Communication Safety \(Apple\) \[13-17yo receiving nudes\]](#)
- #43 [SensitiveContentAnalysis API \(Apple\) \[receiving nudes\]](#)
- #44 [Requiring the recipient to tap on the received image to view it or to unblur it](#)
- #45 [Hide photos on your iPhone, iPad, or Mac with the Hidden album](#)
- #46 [Delivering an Enhanced Privacy Experience in Your Photos App \(Apple\) \[Limited Photos Library\]](#)
- #47 [How to select and view only the albums you want to see in Galaxy gallery](#)
- #48 [Privacy glasses and wearables](#)
- #49 [Phone screen protector for privacy \[phone accessory\]](#)
- #50 [Protecting application-level saved pictures with a password or with biometric authentication](#)
- #51 [Requiring a password or biometric authentication to open the app](#)
- #52 [Encryption \(Android\) \[full-disk or file-based encryption\]](#)
- #53 [iPhone in Business, Security Overview \[weak device encryption\]](#)
- #54 [Apple Platform Security \[strong device encryption\]](#)
- #55 [Erase a device in Find My on iPhone](#)
- #56 [Find, lock, or erase a lost Android device](#)
- #57 [iCloud data security overview \[Standard data protection: "Your iCloud data is encrypted, the encryption keys are secured in Apple data centers"\]](#)
- #58 [About encrypted backups on your iPhone, iPad, or iPod touch \[end-to-end encrypted backups\]](#)
- #59 [Advanced Data Protection for iCloud \[end-to-end encrypted backups\]](#)
- #60 [iCloud data security overview \[end-to-end encrypted photos synchronisation in Advanced Data Protection\]](#)
- #61 [Back up photos & videos \(Android\) \[user-controlled selection of folders of photos to be backed up\]](#)
- #62 [Back up user data with Auto Backup \(Android\) \[end-to-end encryption of backups\]](#)
- #63 [SSL/TLS to encrypt communications between client and server, or to encrypt DNS traffic](#)
- #64 [Anonymity VPNs](#)
- #65 [End-to-end encryption \(sometimes branded as 'secret chat'\)](#)
- #66 [Back up user data with Auto Backup \(Android\) \[XML rules or BackupAgent: excluding specific picture folders from the backup process\]](#)
- #67 [Ad blockers \(service bundled in VPN, or provided with a specific app\) \[if they prevent third party code from loading or using the network\]](#)
- #68 [Screenshot/Recording detection and notification](#)
- #69 [Blocking users](#)
- #70 [Reporting users or reporting messages about abusive content](#)
- #71 [Third party apps for inserting image-based visible watermark onto an existing picture](#)
- #72 [Xiaomi Gallery update brings protective watermark, improved photo organization and more \[text-based visible watermark insertion\]](#)

- #73 Sex Talk (sexual education app)
- #74 iBuddy (privacy education app)
- #75 FoxIT (privacy education app)
- #76 Safety tips and guidelines for users
- #77 Parental supervision: monitoring the linked account of a minor (list of followers, reported users, connection times, etc.)
- #78 Parental supervision: unlocking capabilities at parents' discretion (account type, loosening restrictions for sexual content, receive DMs from non users not followed, etc.)
- #79 Supervision guides for parents and guardians
- #80 Support apps and sites, including peer support apps
- #81 ClickCEOP [child-safety button on Facebook UK]

Appendix G. Overlaps

The following table lists the techniques/approaches we identified to overlap more than one category or threat.

Technique	Overlaps
Certificate pinning	It protects against adversarial Third Parties, and mitigates attacks by adversarial recipients who try to hack app restrictions by doing a PiTM on the own traffic (Adversarial-Recipient Security).
Nude detection algorithms	They can be used both for Reception Control and for Visualisation Privacy.
Recapture detection algorithms, plus systems to certify image ownership or to enforce forwarding/posting policies	They can benefit the privacy of senders, by discouraging or preventing the spread of their pictures (Accountability, and Post Exposure U2U Privacy). They can in turn benefit the privacy of recipients, by preventing intimate interactions with dishonest users (Authenticity).
E2EE	It protects against malicious app servers (App Platform) if the app binary is audited and fingerprints are verified, but it equally protects from network adversaries (Third Parties).
Ephemeral messaging	It provides Post Exposure U2U Privacy, but is also effective against external adversaries who hack the mobile device long after the reception of the nudes (Third Parties).
Ephemeral messaging, when not controlled by the app platform (e.g., ShareIff paper)	Self-destruction schemes may protect against malicious apps (App Platform), in addition to Post Exposure U2U Privacy and Third Parties.
Middleware for image encryption running on TEE (e.g., Rushmore paper)	It mitigates attacks performed by recipients on the own device that exploit an OS vulnerability (attack B3, Adversarial-Recipient Security). Since the app platform cannot access image data, it is also effective to protect user privacy in the face of the App Platform.
Remote attestation, when not controlled by the app platform	It provides security against both adversarial apps (App Platform) and adversarial recipients (Adversarial-Recipient Security).
Tap-to-open picture	It provides Visualisation Privacy, but is also functionally useful in <i>view once</i> to track when the user opens the image (Post Exposure U2U Privacy) and in Reception Control.
Profile information augmentation and purging of malicious profiles	It increases Pre Exposure U2U Privacy, but it may also contribute to Reception Control.
Image information reduction	It works towards Pre Exposure U2U Privacy, but it also helps reduce the privacy impact in the face of adversarial Third Parties and semi-honest App Platforms.
Recapture detection algorithms in Forwarding and Posting Policies for screen-cam attacks	It works equally for malicious recipients in Post Exposure U2U Privacy than for malicious Third Parties.
Hidden, password-protected or otherwise less accessible albums	It contributes to the privacy of sexters when a third party has physical access to the device, as in incident H4. It also contributes to prevent third parties from inadvertently seeing the nudes (incident K2).